# ANOVA

## Multi-Branch Restaurant Enterprise Management System

### A Project Report



# Abstract

ANOVA – Multi-Branch Restaurant Enterprise Management System is a web-based application designed to centralize and streamline the management of restaurant businesses operating across multiple branches. Managing various restaurant activities through manual processes or separate applications can lead to data inconsistencies, inventory shortages, inefficient employee management, delays in order processing, and difficulties in monitoring overall business performance. ANOVA aims to address these challenges by integrating essential restaurant operations into a single, unified platform.

The system incorporates multiple modules, including user and role management, branch management, human resource management, payroll, inventory and procurement, menu management, table reservations, order processing, kitchen operations, billing, customer management, online ordering, delivery coordination, marketing, customer feedback, and business analytics. It supports different categories of users, such as owners, administrators, branch managers, HR staff, inventory managers, chefs, cashiers, employees, suppliers, delivery personnel, and customers. Role-Based Access Control (RBAC) ensures that each user can access only the functionalities and information permitted by their assigned role.

The application will be developed using the MERN stack, comprising MongoDB, Express.js, React.js, and Node.js. React.js will provide an interactive and responsive user interface, while Node.js and Express.js will handle server-side operations and REST API requests. MongoDB, managed through Mongoose, will store and organize application data. JWT-based authentication and secure password hashing will be implemented to protect user accounts and restrict unauthorized access.

ANOVA is intended to improve operational coordination, maintain centralized records, support inventory control, simplify order and billing workflows, and provide branch-wise as well as enterprise-level reports. Payment processing will be simulated, delivery tracking will use sample GPS coordinates, and food recommendations will follow predefined rules. The proposed system aims to provide a secure, maintainable, and scalable foundation for efficient multi-branch restaurant management.

**Keywords:** Multi-Branch Restaurant Management, MERN Stack, Role-Based Access Control, Inventory Management, Human Resource Management, Online Ordering, Business Analytics.

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# Chapter 1: Introduction to Project

## i. Overview

ANOVA – Multi-Branch Restaurant Enterprise Management System is a web-based application designed to simplify and centralize the management of restaurant businesses operating across multiple branches. The system integrates major restaurant activities into a single platform, enabling efficient coordination between administration, human resources, inventory, procurement, kitchen operations, billing, customer services, delivery, and business analytics.

In traditional restaurant management, different departments may rely on manual records or separate applications to handle daily operations. Such practices can lead to data duplication, inventory wastage, employee management difficulties, delays in order processing, and limited visibility into business performance. ANOVA aims to overcome these challenges by providing a centralized database and connected workflows for restaurant operations.

The system supports multiple user roles, including Owner, Administrator, Manager, HR, Inventory Manager, Chef, Cashier, Employee, Supplier, Delivery Personnel, and Customer. Each role is provided with specific functionalities according to its responsibilities. Role-Based Access Control (RBAC) helps ensure that users can access only the pages, records, and operations authorized for their assigned roles.

The application will be developed using the MERN stack, which consists of MongoDB, Express.js, React.js, and Node.js. React.js will provide the frontend interface, Express.js and Node.js will handle backend processing and REST APIs, and MongoDB will store application data. Mongoose will support database schemas, validation, and data relationships, while JWT-based authentication and password hashing will help secure user accounts.

The major modules of ANOVA are:

**Authentication and Authorization Module**

This module manages user authentication and controls access to protected application features. It ensures that users are identified before accessing restricted resources and that permissions are checked on the server.

Functions:

- User login and credential verification.
- Secure password hashing and verification.
- JWT-based authentication.
- Role-based authorization.
- Protected frontend routes and backend API endpoints.
- User account status and permission validation.

**User and Role Management Module**

This module allows authorized administrators to manage user accounts, roles, permissions, and branch assignments. It helps maintain a structured access system across the restaurant enterprise.

Functions:

- Create, update, and manage user accounts.
- Assign appropriate roles to users.
- Configure permitted access rights.
- Activate or deactivate user accounts.
- Manage branch and departmental assignments.
- Enforce restrictions on sensitive operations.

**Multi-Branch Management Module**

This module provides centralized management of restaurant branches and maintains branch-specific information.

Functions:

- Create and update branch records.
- Maintain branch addresses and contact information.
- Manage operating hours and branch status.
- Assign branch managers and employees.

- Associate orders, inventory, and operational records with branches.

**Dashboard and Administration Module**

This module presents relevant business information according to the user's role. It helps owners, administrators, and branch managers monitor operations and make informed decisions.

Functions:

- Display role-specific dashboards.
- Show sales, revenue, and order summaries.
- Display branch-wise performance indicators.
- Monitor inventory and operational summaries.
- Present charts and key performance indicators.
- Provide access to authorized administrative settings.

**Employee and Human Resource Management Module**

This module manages employee information and workforce-related activities. It helps HR staff and authorized managers maintain organized employee records across different branches.

Functions:

- Create and maintain employee profiles.
- Record job roles and employment details.
- Manage employee branch assignments.
- Create and maintain work shifts.
- Record and review attendance.
- Process leaves applications and approvals.

**Payroll Management Module**

The payroll module manages salary-related information and maintains payroll records for employees. Access to salary details and payroll operations is restricted to authorized users.

Functions:

- Maintain salary structures.
- Calculate salaries using configured rules.
- Record applicable deductions.
- Generate payroll records.
- Maintain payroll history.
- Generate salary summaries and reports.

- Restrict access to confidential payroll information.

**Audit Log Module**

This module maintains records of important administrative and system activities. It supports accountability by recording relevant actions performed by authorized users.

Functions:

- Record important administrative events.
- Store the actor, action, resource, and timestamp.
- Maintain relevant event details and status.
- Filter and paginate audit records.
- Allow authorized users to review system activity.

**Inventory Management Module**

This module tracks ingredients and stock quantities at individual restaurant branches. It helps inventory managers monitor stock availability, identify shortages, and reduce wastage.

Functions:

- Create and maintain ingredient records.
- Manage units of measurement and stock quantities.
- Monitor low-stock conditions.
- Track expiry dates.
- Record stock additions and adjustments.
- Maintain inventory movement history.
- Generate inventory and valuation reports.

**Supplier and Purchase Management Module**

This module manages supplier information and procurement activities. It connects inventory requirements with purchase orders and the recording of received goods.

Functions:

- Maintain supplier profiles and contact details.
- Associate suppliers with relevant ingredients.
- Create purchase orders.
- Track purchase order statuses.
- Record received goods and quantities.
- Maintain procurement history.

- Monitor pending purchasing requirements.

**Stock Movement and Wastage Management Module**

This module records the movement and loss of inventory between branches and within restaurant operations. It helps maintain accurate stock information and identify avoidable wastage.

Functions:

- Manage interbranch stock transfers.
- Record ingredient consumption.
- Record damaged and expired stock.
- Maintain wastage quantities and reasons.
- Update inventory after authorized stock movements.
- Generate stock movement and wastage reports.

**Menu Management Module**

This module allows authorized staff to manage food items and their availability across restaurant branches.

Functions:

- Create and manage menu categories.
- Add and update food items.
- Maintain descriptions, images, and prices.
- Store recipe information where applicable.
- Update food item availability.
- Display available menu items to customers.

**Table and Reservation Management Module**

This module manages restaurant tables and customer reservations. It helps restaurant staff organize seating arrangements and maintain reservation records.

Functions:

- Register tables and seating capacities.
- Monitor table availability.
- Create customer reservations.
- Confirm or cancel reservations.
- Allocate tables according to business rules.

- Maintain reservation history and status.

**Order Management Module**

This module manages orders received through dine-in, takeaway, and online ordering channels. It connects customer requests with kitchen preparation, billing, and delivery operations.

Functions:

- Create and manage customer orders.
- Support dine-in, takeaway, and online orders.
- Add items and quantities to orders.
- Track order status throughout processing.
- Associate orders with customers and branches.
- Maintain order history.
- Coordinate order information with kitchen, billing, and delivery modules.

**Kitchen Workflow Module**

This module supports the preparation of food orders by kitchen staff. It provides visibility into incoming orders and enables staff to communicate preparation progress to other departments.

Functions:

- Display incoming food orders.
- Maintain a kitchen preparation queue.
- Show ordered items and preparation details.
- Update preparation statuses.
- Prioritize orders where required.
- Mark orders as ready for service or pickup.

**Billing and Payment Module**

This module manages bill generation and payment records for eligible orders. It helps cashiers calculate payable amounts and maintain transaction information.

Functions:

- Retrieve eligible orders for billing.
- Generate customer bills.
- Calculate applicable taxes and discounts.
- Record payment status.
- Generate receipts.

- Maintain billing and transaction history.

**Food Recommendation Module**

This module suggests food items to customers using predefined rules and available ordering information. It aims to improve menu discovery without requiring a complex machine-learning system.

Functions:

- Recommend items based on previous orders.
- Suggest dishes according to recorded preferences.
- Display popular food items.
- Suggest frequently ordered dishes.
- Apply predefined recommendation rules.

**Customer Management Module**

This module maintains customer profiles and relevant interaction information. It supports personalized service and provides the information required for ordering, delivery, and customer history.

Functions:

- Create and update customer profiles.
- Maintain contact information.
- Manage delivery addresses.
- Store relevant preferences where applicable.
- Maintain customer order history.

**Online Ordering Module**

This module allows customers to browse menus and place orders through the web application. It connects the customer interface with order processing, checkout, payment status, and order tracking.

Functions:

- Browse restaurants and available menus.
- Add items to a shopping cart.
- Change item quantities or remove items.
- Calculate the order total.
- Apply eligible coupons and promotions.

- Complete checkout and confirm orders.

**Delivery Management and Tracking Module**

This module coordinates delivery assignments and tracks delivery progress. It allows delivery personnel to update order statuses and provides customers with delivery progress information.

Functions:

- Assign delivery orders to delivery personnel.
- Display assigned delivery details.
- Provide necessary customer delivery information.
- Update pickup and delivery statuses.
- Display simulated GPS-based delivery tracking using sample coordinates.
- Maintain delivery history.
- Generate delivery performance reports.

**Marketing and Promotions Module**

This module manages promotional offers and discount coupons for eligible customers. It supports marketing activities while applying predefined conditions to prevent invalid coupon usage.

Functions:

- Create and manage promotional campaigns.
- Generate discount coupons.
- Configure offer validity periods.
- Define eligibility and discount rules.
- Validate coupons during checkout.
- Display active offers to customers.

**Customer Service and Feedback Module**

This module enables customers to share their experiences and raise complaints. It helps authorized staff review feedback and maintain records of customer concerns.

Functions:

- Submit customer ratings and reviews.
- Collect customer feedback.
- Register complaints.
- Track complaint status.

- Review customer concerns through authorized staff.

**Reports and Analytics Module**

This module converts operational data into reports and visual summaries. It helps owners, administrators, and branch managers evaluate business performance and identify areas requiring attention.

Functions:

- Generate branch-wise sales and revenue reports.
- Summarize order volumes and statuses.
- Generate inventory and wastage reports.
- Provide employee and HR summaries.
- Generate delivery performance reports.
- Display charts, filters, and key performance indicators.
- Support enterprise-level and branch-level performance monitoring.

Through these interconnected modules, ANOVA aims to provide a unified environment for managing restaurant operations, improving coordination between departments, and maintaining consistent information across multiple branches.

## ii. Project Objective

The primary objective of ANOVA is to design and develop a secure, centralized, and role-based web application that integrates the major operational activities of a multi-branch restaurant enterprise.

The project aims to reduce dependence on disconnected or manual processes by connecting administrative, employee, inventory, ordering, billing, and customer service operations through a common platform.

The specific objectives are as follows:

1. *Centralized Management:* To maintain a unified system for managing multiple restaurant branches and their operational records.
2. *Secure Access Control:* To implement authentication and Role-Based Access Control (RBAC) so that users can access only the functionalities and information permitted by their roles.
3. *User and Branch Administration:* To manage user accounts, role assignments, branch

records, and employee assignments efficiently.

4. *Human Resource Management:* To maintain employee profiles, shifts, attendance, leave applications, and payroll records.
5. *Inventory Control:* To monitor ingredient quantities, expiry dates, stock movements, and wastage to improve inventory visibility and reduce avoidable losses.
6. *Procurement Management:* To organize supplier information, purchase orders, received goods, and purchasing history.
7. *Restaurant Operations:* To manage menus, table reservations, customer orders, and kitchen preparation workflows.
8. *Billing and Transactions:* To generate bills, calculate applicable taxes and discounts, maintain receipts, and record payment statuses.
9. *Customer Convenience:* To provide customers with menu browsing, online ordering, checkout, reservations, order history, and order tracking.
10. *Delivery Coordination:* To assign deliveries, update delivery statuses, and demonstrate delivery tracking using sample GPS coordinates.
11. *Customer Engagement:* To support food recommendations, promotional offers, ratings, reviews, feedback, and complaint management.
12. *Business Analytics:* To generate branch-wise and enterprise-level reports related to sales, orders, inventory, employees, and deliveries.
13. *Operational Accountability:* To maintain audit logs of important administrative activities and support authorized review of system operations.
14. *Maintainable Application Design:* To develop a modular web application using the MERN stack, with reusable frontend components and organized backend services.
15. *Responsive User Interface:* To provide a consistent and user-friendly interface that supports common desktop, tablet, and mobile screen sizes.

The overall objective is to establish a structured digital platform that improves information accessibility, supports coordinated restaurant workflows, and provides management with a clearer view of business operations.

## iii. Scope of the Project

The scope of ANOVA covers the design and development of a web-based system for managing the major activities of a restaurant enterprise operating across multiple branches. The application will integrate administration, workforce management, inventory, procurement, restaurant operations, billing, customer services, delivery coordination, marketing, and analytics.

The system will provide separate interfaces and permissions for internal users, external suppliers, delivery personnel, and customers. Authorized users will be able to access information according to their roles and relevant branch assignments.

The major areas covered within the project scope are:

**Addressing the Time Constraints**

The development of a complete restaurant enterprise management system involves several interconnected modules. Developing every module with advanced automation and external integrations within a limited academic project duration may not be feasible. Therefore, ANOVA will follow a modular and prioritized development approach to complete the essential functionality within the available time.

The following measures will be adopted:

- *Modular Development:* The application will be divided into independent modules, allowing each part to be designed, implemented, and tested systematically.
- *Prioritization of Core Features:* Authentication, role management, branch management, inventory, orders, billing, and other essential workflows will receive priority.
- *Reusable Components:* Common frontend components, form elements, tables, validation logic, and backend middleware will be reused to reduce development effort.
- *Use of the MERN Stack:* React.js, Express.js, Node.js, MongoDB, and Mongoose will provide a consistent development environment for implementing the application.
- *REST API Structure:* Standardized API endpoints will simplify communication

between the frontend and backend and support parallel development of modules.

- *Defined Project Boundaries:* Payment processing will be simulated, delivery tracking will use sample GPS coordinates, and recommendations will use predefined rules rather than complex external services.
- *Incremental Integration:* Modules will be integrated progressively to identify compatibility problems before final testing.
- *Planned Testing:* Functional, integration, and authorization testing will be performed throughout development rather than being postponed until the final stage.

These measures will help control development complexity and make the project achievable within the available academic schedule. However, the completion of all proposed modules will depend on appropriate task allocation, implementation priorities, and regular testing.

**Addressing the Performance Constraint**

ANOVA must provide responsive interaction and efficient data access even when multiple branches, employees, menu items, orders, and inventory records are maintained. Performance constraints will be addressed through suitable application design, database optimization, and efficient request handling.

The following measures will be adopted:

- *Efficient Database Queries:* MongoDB queries will retrieve only the data required for the requested operation.
- *Database Indexing:* Appropriate indexes will be created for frequently searched fields, such as user identifiers, branch references, order statuses, and relevant timestamps.
- *Pagination:* Large lists, including orders, employees, inventory records, and audit logs, will be retrieved in manageable pages.
- *Backend Validation:* Requests will be validated before processing to prevent invalid operations and unnecessary database work.
- *Reusable API Middleware:* Authentication, authorization, and common validation logic will be organized into reusable middleware.
- *Role and Branch Filtering:* Queries will restrict results to the records that the user is authorized to access, avoiding unnecessary retrieval of unrelated data.

- *Responsive Frontend Design:* React components will be structured to support smooth navigation and efficient rendering of frequently used screens.
- *Controlled Data Retrieval:* Dashboards and reports will request relevant summaries rather than loading all underlying records at once.
- *Error Handling:* Consistent error handling will help prevent failed operations from disrupting normal application workflows.
- *Performance Testing:* API response times, database queries, and important user workflows will be tested to identify bottlenecks.

These measures are intended to improve responsiveness and maintainability. Actual performance will depend on the deployment environment, database size, query patterns, network conditions, and number of concurrent users; therefore, performance improvements will be evaluated through testing rather than assumed in advance.

**Overall Project Scope**

The completed application is expected to support centralized restaurant management, role-specific access, connected operational workflows, and consolidated reporting. Its intended scope includes the following capabilities:

i. Management of multiple restaurant branches.
ii. Authentication, user administration, and role-based authorization.
iii. Employee records, attendance, shifts, leave, and payroll.
iv. Ingredient inventory, suppliers, procurement, stock transfers, and wastage.
v. Menu management, table reservations, order processing, and kitchen workflows.
vi. Billing, receipts, and simulated payment processing.
vii. Customer profiles, online ordering, and order tracking.
viii. Delivery assignments and simulated GPS-based tracking.
ix. Rule-based food recommendations.
x. Promotions, coupons, reviews, ratings, feedback, and complaints.
xi. Audit logs and business analytics.

# Chapter 2: System Analysis and Design

## i. Disadvantages of Traditional Process

Traditional restaurant management processes often depend on manual registers, spreadsheets, phone calls, and separate software applications to manage daily activities. When a restaurant business operates across multiple branches, these methods can make it difficult to maintain consistent information, coordinate departments, and monitor overall business performance.

The major disadvantages of traditional restaurant management processes are:

- Manual Record Maintenance
- Lack of Centralized Management
- Inefficient Inventory Management
- Difficult Employee and Payroll Management
- Delays in Order Processing
- Inefficient Billing and Payment Records
- Limited Customer Convenience
- Poor Coordination with Suppliers and Delivery Personnel
- Limited Reporting and Business Analysis
- Security and Access Control Issues

Therefore, a centralized and role-based system is required to reduce these limitations and improve the coordination, accuracy, and accessibility of restaurant management information.

## ii. Proposed System

ANOVA – Multi-Branch Restaurant Enterprise Management System is proposed as a centralized web-based solution for managing the major operations of restaurant businesses across multiple branches. The system integrates administrative activities, employee management, inventory, procurement, restaurant operations, billing, customer services, delivery, marketing, and reporting into a unified application.

The proposed system will provide role-specific interfaces for owners, administrators, branch

managers, HR staff, inventory managers, chefs, cashiers, employees, suppliers, delivery personnel, and customers. Each user will be granted access according to their assigned role and applicable branch permissions.

The application will be developed using the MERN stack, consisting of MongoDB, Express.js, React.js, and Node.js. The frontend will provide interactive pages and dashboards, while the backend will process requests, apply business rules, validate data, and enforce authorization. MongoDB will maintain centralized application records, and Mongoose will support schema validation and data relationships.

The proposed system will connect related workflows. For example, customer orders will be forwarded to the kitchen for preparation, eligible orders will be processed through billing, and delivery orders will be assigned to delivery personnel. Similarly, inventory and procurement records will help authorized staff monitor stock levels and purchasing requirements.
The system will also support consolidated and branch-wise reports, audit logs, customer feedback, promotional offers, and rule-based food recommendations. Payment processing and GPS-based delivery tracking will be simulated for project demonstration purposes.

**Benefits**

The proposed ANOVA system offers the following benefits:

- Centralized Information Management
- Improved Operational Efficiency
- Better Inventory Control
- Simplified Employee Management
- Secure Access Control
- Improved Customer Experience
- Organized Billing and Transactions
- Better Delivery Coordination
- Improved Reporting and Decision Support
- Improved Maintainability and Extensibility
- Reduced Administrative Effort

## iii. Requirements Gathering Methods

Requirements gathering is the process of identifying and documenting the functional and non-functional needs of a proposed software system. It helps determine what the application must do, who will use it, and what constraints must be considered during development.

For ANOVA, requirements gathering focuses on understanding the responsibilities of different restaurant stakeholders and identifying the workflows required to manage multiple branches through a centralized platform.

The following methods can be used to identify and organize the project requirements.

**Stakeholder Identification**

The first step is to identify the users and stakeholders who interact with the system or depend on its information.

- Owners require enterprise-level business reports.
- Administrators require user, role, and branch management facilities.
- Branch managers require branch-level operational controls.
- HR staff require employee, attendance, leave, and payroll facilities.
- Inventory managers require stock, supplier, and procurement functions.
- Kitchen staff require incoming order and preparation information.
- Cashiers require billing and payment record functions.
- Employees require access to assigned operational tasks.
- Suppliers require relevant purchase order information.
- Delivery personnel require assigned delivery details.
- Customers require menu browsing, reservations, ordering, and feedback facilities.

Identifying these stakeholders helps ensure that the application covers the major responsibilities associated with restaurant operations.

**Study of Existing Processes**

The existing restaurant workflows can be studied to identify problems associated with manual records and disconnected applications.

- Examine how employee information and attendance are maintained.
- Understand how inventory levels and purchases are recorded.
- Study how orders are communicated to kitchen staff.
- Review the billing and payment recording process.
- Identify how reservations and delivery assignments are managed.
- Examine how branch-wise reports are prepared.

This study helps identify repetitive tasks, possible errors, delays, and opportunities for workflow improvement.

**Use-Case Identification**

Use cases describe the interactions between users and the system to achieve specific objectives. They help translate stakeholder responsibilities into application functionality.

Examples include:

- Owner views consolidated business reports.
- Administrator manages user accounts and branches.
- HR staff records employee attendance.
- Inventory manager creates a purchase order.
- Chef updates an order's preparation status.
- Cashier generates a bill.
- Customer places an online order.
- Delivery personnel update delivery status.

Use-case identification helps define the system boundary and the expected interactions between actors and application modules.

**Functional Decomposition**

The overall system is divided into smaller modules to make requirements easier to understand, implement, and test.

- Authentication and authorization.
- User and branch management.
- HR and payroll.
- Inventory and procurement.
- Menu and reservation management.
- Orders and kitchen operations.
- Billing and payments.
- Customer services and delivery.
- Marketing and analytics.

**Validation of Requirements**

The identified requirements should be reviewed for clarity, consistency, feasibility, and testability before implementation.

- Check whether each requirement has a clear purpose.
- Confirm that user roles have appropriate permissions.
- Verify that related modules exchange the necessary information.
- Identify conflicting or duplicated requirements.
- Define expected outcomes for important workflows.
- Use functional and authorization tests to validate implemented requirements.

These methods provide a structured basis for defining the requirements and designing the proposed system. Any stakeholder interviews or questionnaires should be documented separately if they are actually conducted as part of the project.

## iv. Functional Requirements

Functional requirements specify the operations and services that the system must provide. They describe the expected behaviour of ANOVA when users perform actions through the application.

The major functional requirements are listed below.

**FR-01: Authentication and Authorization**

- The system shall allow registered users to log in using valid credentials.
- The system shall verify credentials securely.
- The system shall use JWT-based authentication for protected requests.
- The system shall restrict access according to assigned roles and permissions.
- The system shall prevent unauthorized users from accessing protected APIs.

**FR-02: User and Role Management**

- The system shall allow authorized administrators to manage user accounts.
- The system shall support role assignment and account status management.
- The system shall associate users with relevant branches where applicable.
- The system shall enforce restrictions on role and permission changes.

**FR-03: Branch Management**

- The system shall allow authorized users to create and update branch records.
- The system shall maintain branch addresses, contact information, operating hours, and status.
- The system shall associate relevant employees and operational records with branches.
- The system shall support branch-wise information retrieval.

**FR-04: Dashboard and Administration**

- The system shall display dashboards according to user roles.
- The system shall present relevant business and operational summaries.
- The system shall provide authorized access to reports and administrative settings.
- The system shall support branch-wise and enterprise-level monitoring.

**FR-05: Employee and HR Management**

- The system shall maintain employee profiles and job details.
- The system shall support employee branch and role assignments.
- The system shall allow authorized HR users to manage shifts and attendance.
- The system shall support leave applications and approval workflows.
- The system shall provide employee and HR summaries.

**FR-06: Payroll Management**

- The system shall maintain employee salary information.
- The system shall calculate salaries using configured rules.
- The system shall record applicable deductions and payroll details.
- The system shall maintain payroll history and summaries.
- The system shall restrict payroll access to authorized users.

**FR-07: Audit Logging**

- The system shall record important administrative activities.
- The system shall maintain relevant actor, action, resource, and timestamp information.
- The system shall allow authorized users to review audit records.
- The system shall support filtering and pagination of audit logs.

**FR-08: Inventory Management**

- The system shall maintain ingredient records and stock quantities.
- The system shall support branch-wise inventory management.
- The system shall track low-stock conditions and expiry dates.
- The system shall record stock adjustments and movements.
- The system shall generate inventory and wastage reports.

**FR-09: Supplier and Purchase Management**

- The system shall maintain supplier information.
- The system shall support supplier-ingredient associations.
- The system shall allow authorized users to create purchase orders.
- The system shall track purchase order status.
- The system shall record received goods and quantities.

**FR-10: Stock Movement and Wastage**

- The system shall record interbranch stock transfers.
- The system shall maintain ingredient consumption records.
- The system shall record damaged and expired stock.
- The system shall store wastage quantities and reasons.
- The system shall maintain inventory movement history.

**FR-11: Menu Management**

- The system shall support the creation and maintenance of menu categories.
- The system shall maintain item descriptions, images, and prices.
- The system shall support recipe information where applicable.
- The system shall allow authorized users to update menu availability.
- The system shall display available menu items to customers.

**FR-12: Table and Reservation Management**

- The system shall maintain restaurant table information and seating capacity.
- The system shall support reservation creation and cancellation.
- The system shall maintain reservation confirmation and status.
- The system shall check table availability according to defined business rules.
- The system shall maintain reservation history.

**FR-13: Order Management**

- The system shall support dine-in, takeaway, and online orders.
- The system shall maintain order items, quantities, and relevant details.
- The system shall associate orders with customers and branches where applicable.
- The system shall track order status.
- The system shall maintain order history and coordinate with related modules.

**FR-14: Kitchen Workflow**

- The system shall display incoming food orders to authorized kitchen staff.
- The system shall provide order items and preparation details.
- The system shall support kitchen preparation status updates.
- The system shall allow eligible orders to be marked as ready for service or pickup.

**FR-15: Billing and Payments**

- The system shall generate bills for eligible orders.
- The system shall calculate applicable taxes and discounts.
- The system shall maintain payment status and transaction records.
- The system shall generate receipts.
- The system shall maintain billing history.
- The system shall support simulated payment processing for demonstration.

**FR-16: Food Recommendations**

- The system shall provide food recommendations using predefined rules.
- The system shall support suggestions based on previous orders or recorded preferences where available.
- The system shall display popular or frequently ordered dishes where relevant.

**FR-17: Customer Management**

- The system shall allow customers to register and manage their profiles.
- The system shall maintain customer contact information.
- The system shall support delivery address management.
- The system shall maintain relevant customer order history.
- The system shall restrict access to customer information according to permissions.

**FR-18: Online Ordering**

- The system shall allow customers to browse restaurants and menus.
- The system shall support adding, updating, and removing cart items.
- The system shall calculate order totals.
- The system shall support checkout and order confirmation.
- The system shall allow customers to view order history and status.

**FR-19: Delivery Management**

- The system shall support delivery assignment to delivery personnel.
- The system shall allow delivery personnel to view assigned orders.
- The system shall support pickup and delivery status updates.

**FR-20: Marketing and Promotions**

- The system shall allow authorized users to manage promotional campaigns.
- The system shall support coupon creation and validity rules.
- The system shall validate coupon eligibility during checkout.
- The system shall display applicable promotions to customers.

**FR-21: Customer Service and Feedback**

- The system shall allow customers to submit ratings and reviews.
- The system shall support customer feedback submission.
- The system shall allow complaints to be registered and tracked.
- The system shall support authorized review of customer concerns.
- The system shall maintain relevant feedback and complaint records.

**FR-22: Reports and Analytics**

- The system shall generate sales and revenue reports.
- The system shall provide order volume and order status summaries.
- The system shall generate inventory and wastage reports.
- The system shall provide employee and HR summaries to authorized users.
- The system shall support delivery performance reporting.
- The system shall display relevant charts and key performance indicators.
- The system shall enforce access restrictions on reports containing sensitive information.

**FR-23: Branch-Level Data Restrictions**

- The system shall restrict branch-specific records according to user permissions.
- The system shall prevent users from modifying records belonging to unauthorized branches.
- The system shall enforce branch-level restrictions on the backend.
- The system shall allow enterprise-level access only to appropriately authorized roles.

## v. Non-Functional Requirements

Non-functional requirements define the quality attributes, constraints, and operational standards that the system should satisfy. They describe how effectively ANOVA must operate rather than specifying individual business operations.

- Security

The system must protect user accounts, confidential information, and restricted operations.

- Performance

The system should respond efficiently to user actions and database requests.

- Reliability

The system should perform its intended operations consistently and handle failures appropriately.

- Usability

The application should be understandable and convenient for users with different responsibilities.

- Maintainability

The system should be organized so that errors can be corrected and features can be modified efficiently.

- Scalability

The system should provide a foundation for supporting growth in restaurant operations.

- Availability

The system should be accessible to authorized users when the application and its supporting services are operational.

- Data Integrity and Consistency

The system must maintain accurate and consistent records across related modules.

- Portability and Responsiveness

The frontend should support common devices and modern web browsers.

- Compatibility and Integration

The frontend, backend, and database must communicate correctly through defined interfaces.

- Testability

The system should support systematic verification of its functionality and quality attributes.

- Data Privacy

The system must limit the exposure of personal and confidential information.

## vi. UML Diagrams

Unified Modeling Language (UML) is a standardized modeling language used in software engineering to visually represent a system's design. UML diagrams provide a way to communicate and document various aspects of a system in a clear and standardized manner. There are several types of UML diagrams, each serving a specific purpose in depicting different facets of a software system.

### Use Case

Describes the interactions between external actors (users or other systems) and the system under consideration.

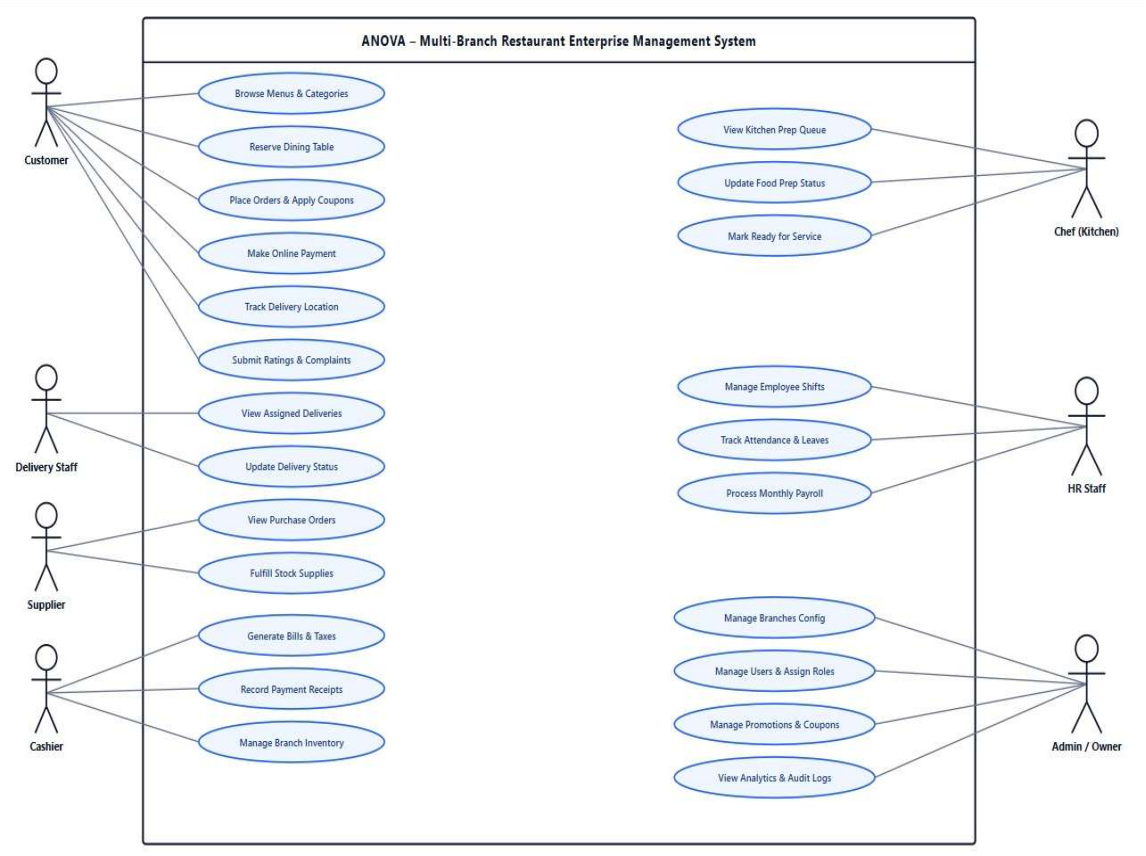



*Figure 1: Use Case Diagram*

**Activity Diagram**

Models the workflow or business processes within a system, depicting the flow of activities and decisions.

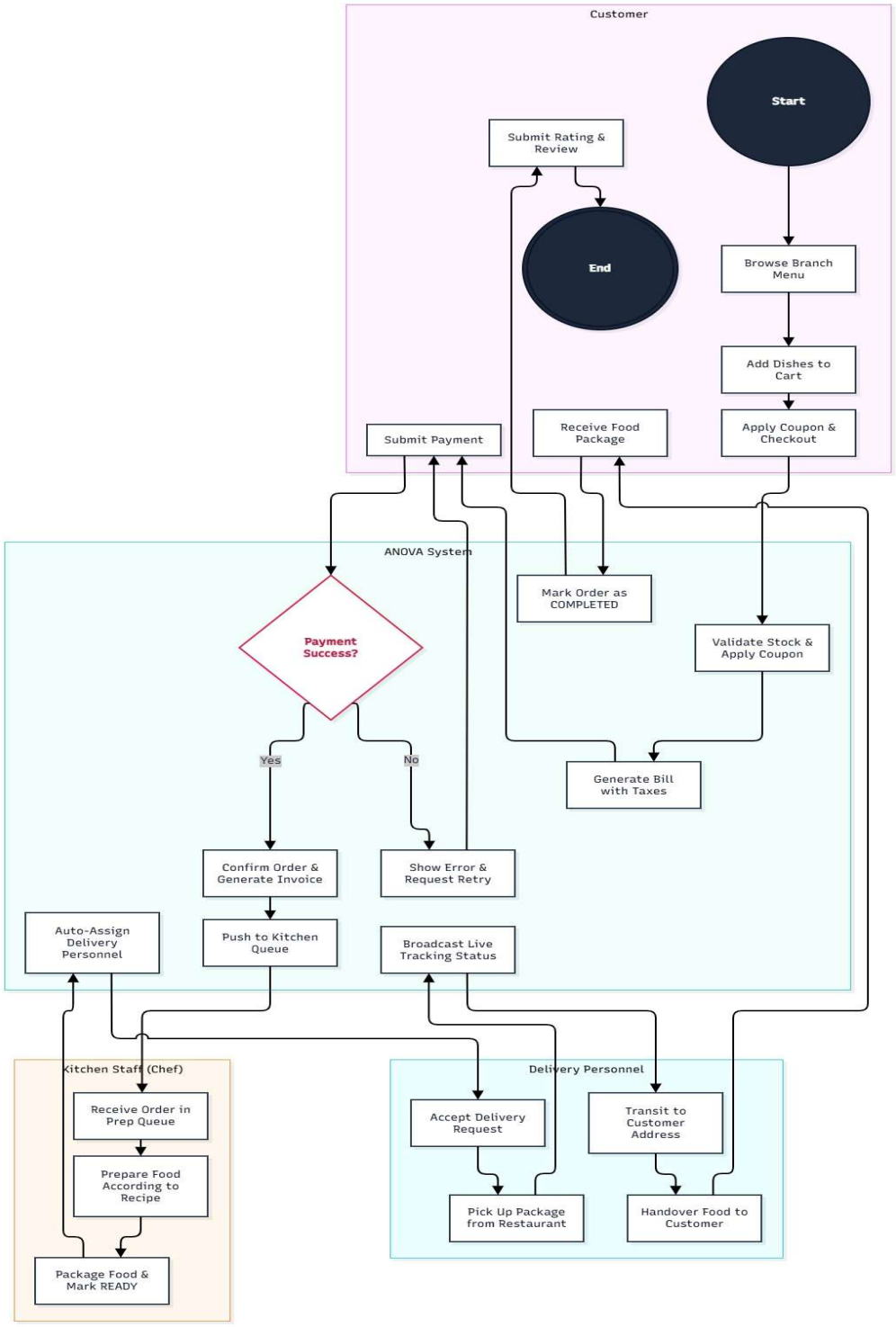



*Figure 2: Activity Diagram*

**Sequence Diagram**

Illustrates the interactions between different objects or components over time,

showcasing the order of messages exchanged.

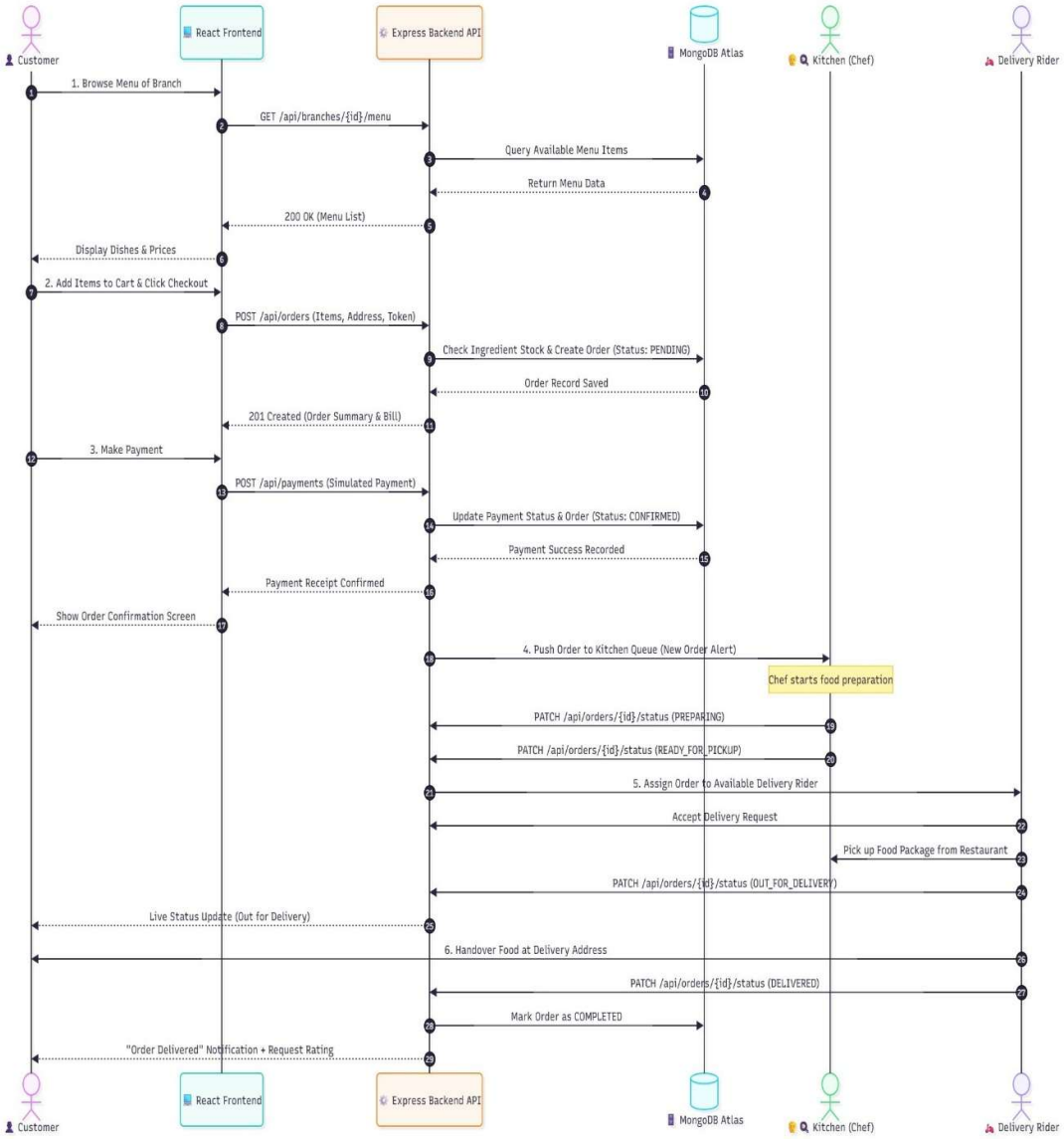



*Figure 3: Sequence Diagram*

**Entity-Relationship (ER) diagram**

Entity-Relationship (ER) diagram is a visual representation of the data model for a

system, illustrating entities, their attributes, and the relationships between entities.

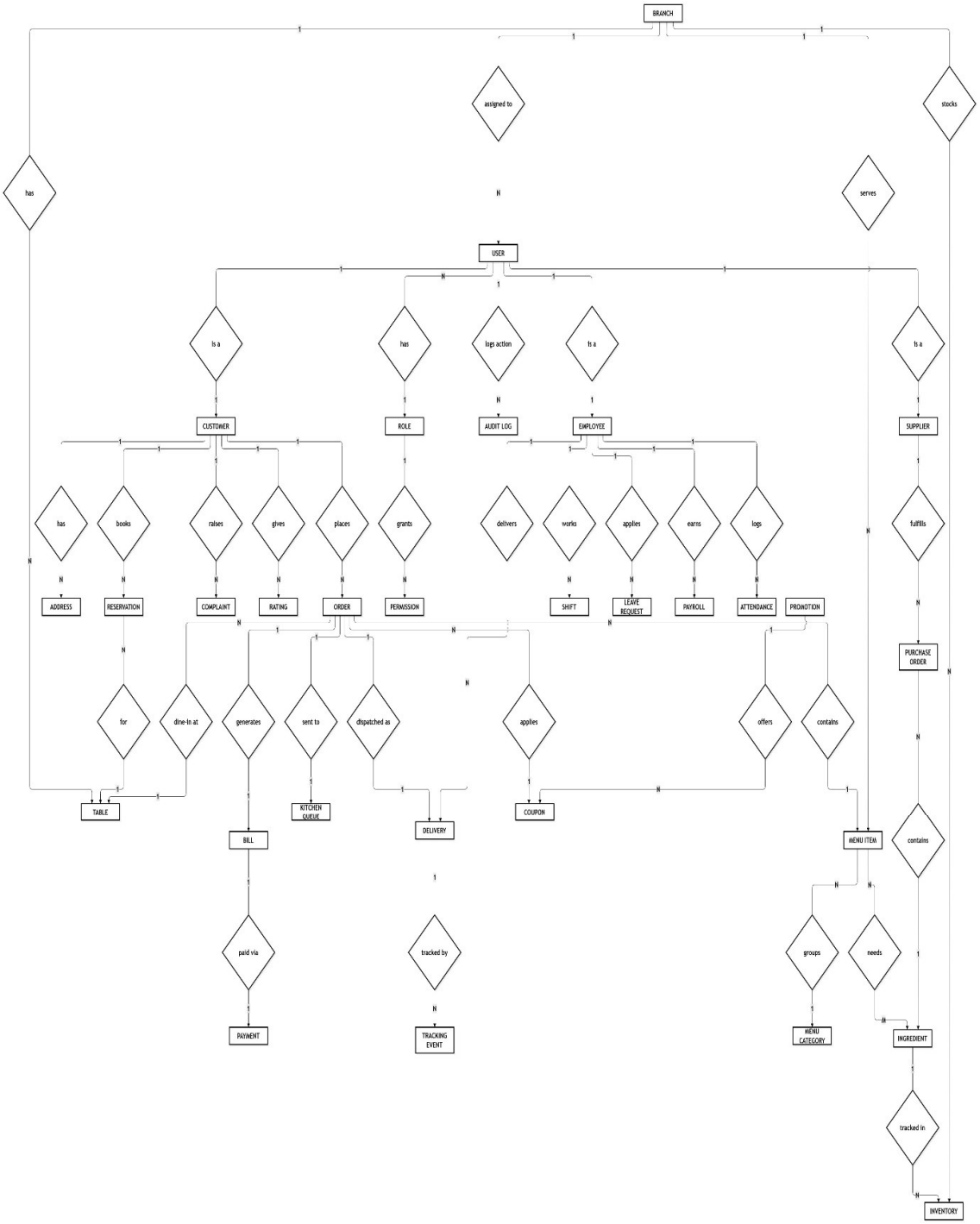



*Figure 4: E-R Diagram*

**Data Flow Diagram (DFD)**

Data Flow Diagram (DFD) is a graphical representation of the flow of data within a system. It illustrates how data moves between processes, data stores, and external entities.

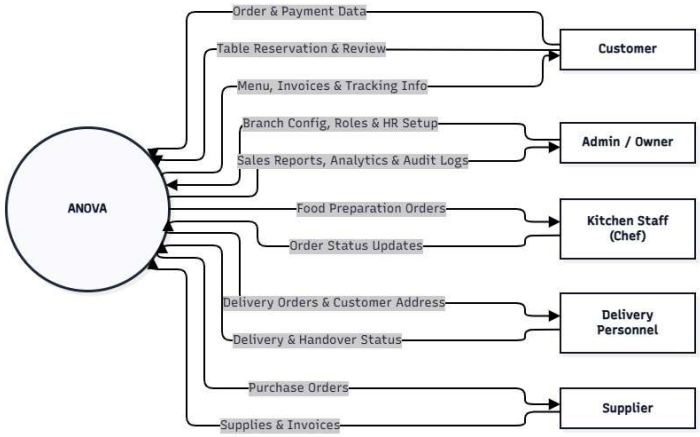



*Figure 5: DFD Level – 0*

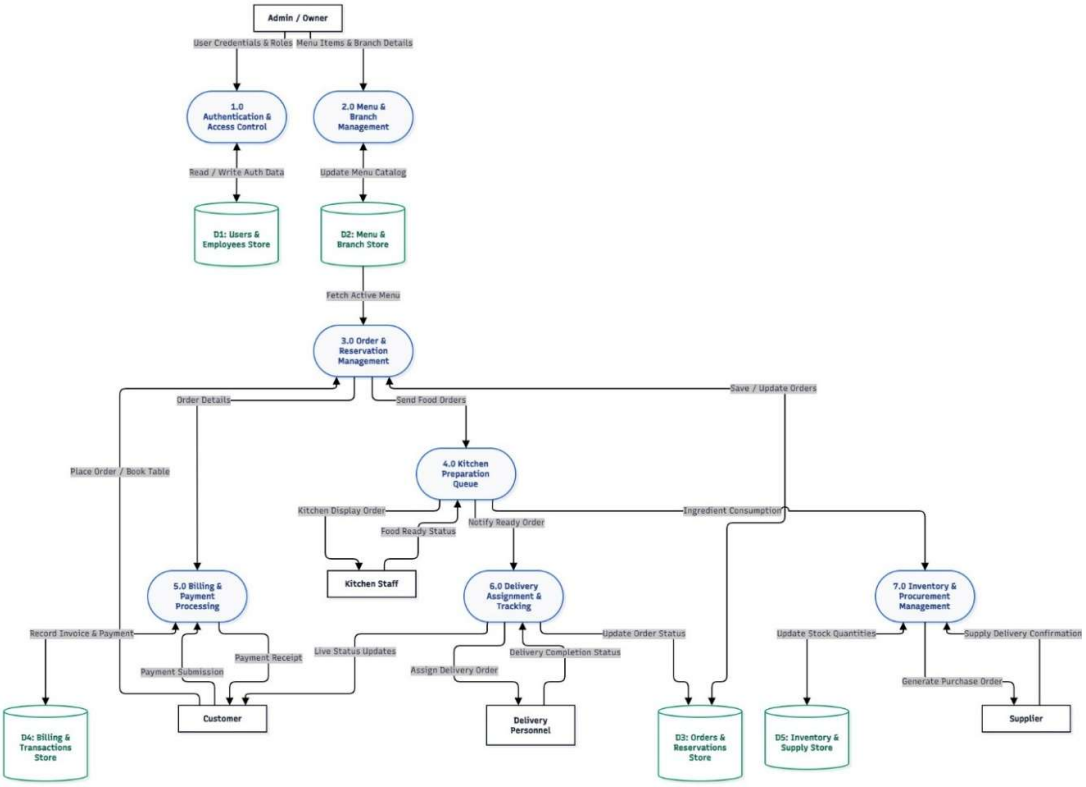



*Figure 6: DFD Level - 1*

# Chapter 3: Implementation of Project

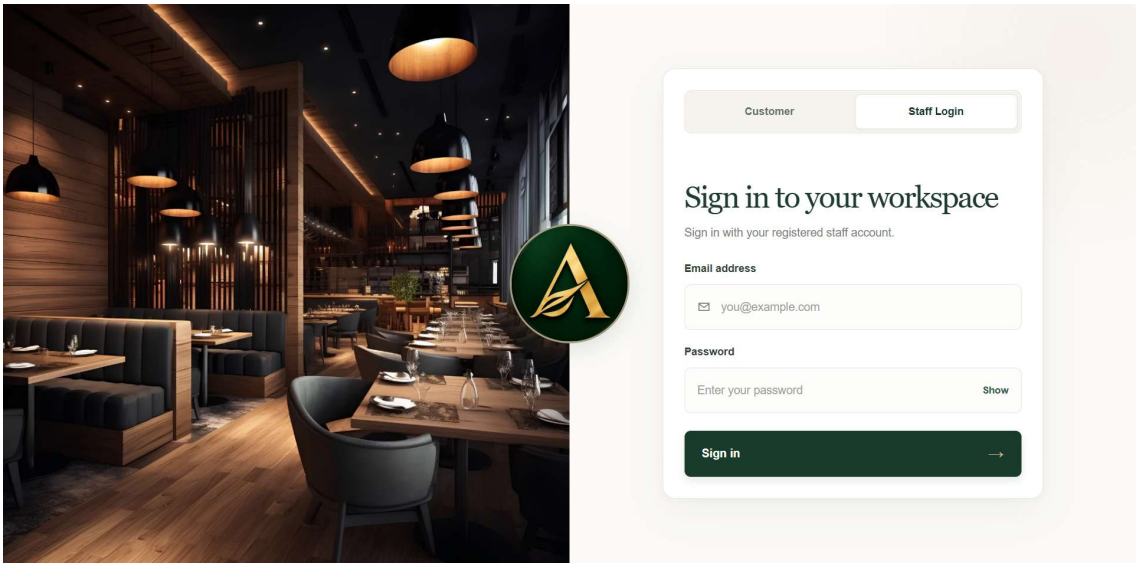



*Figure 7: Login Page Staff Login*

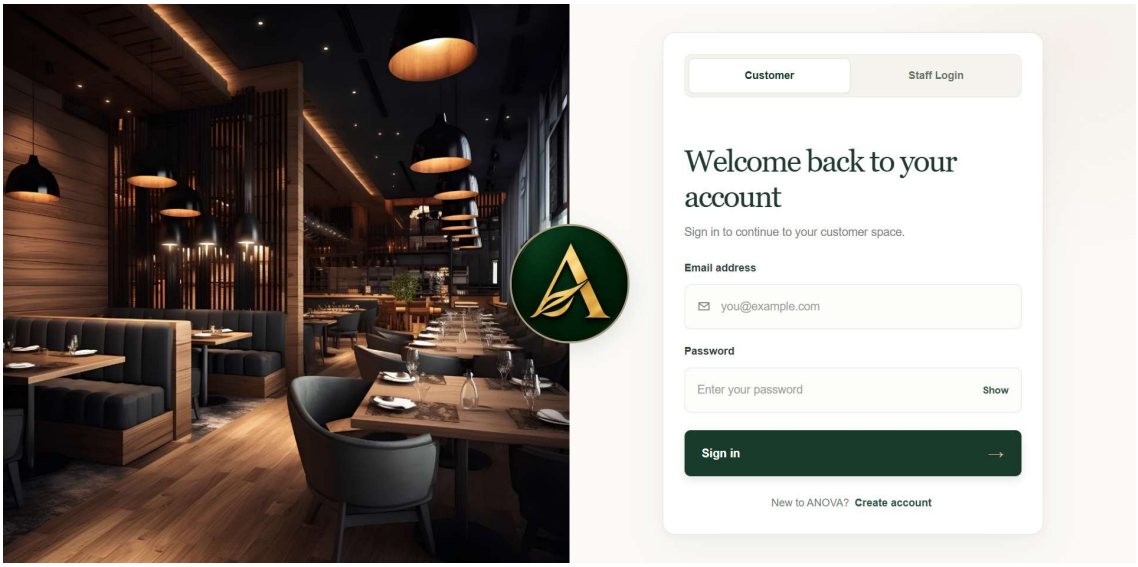



*Figure 8: Login Page Customer Login*

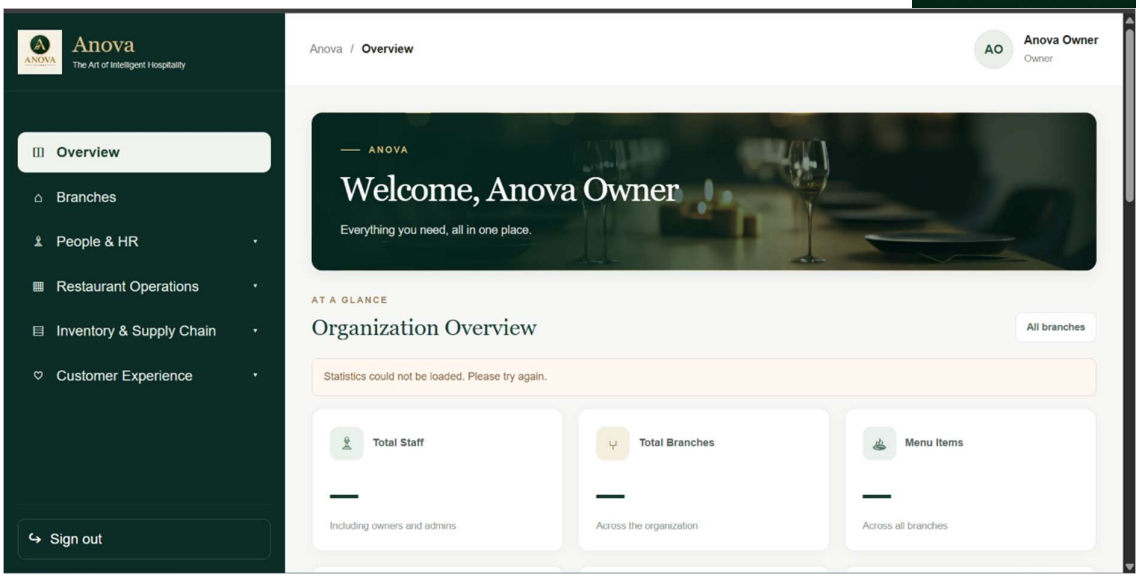



*Figure 9: Overview Dashboard*

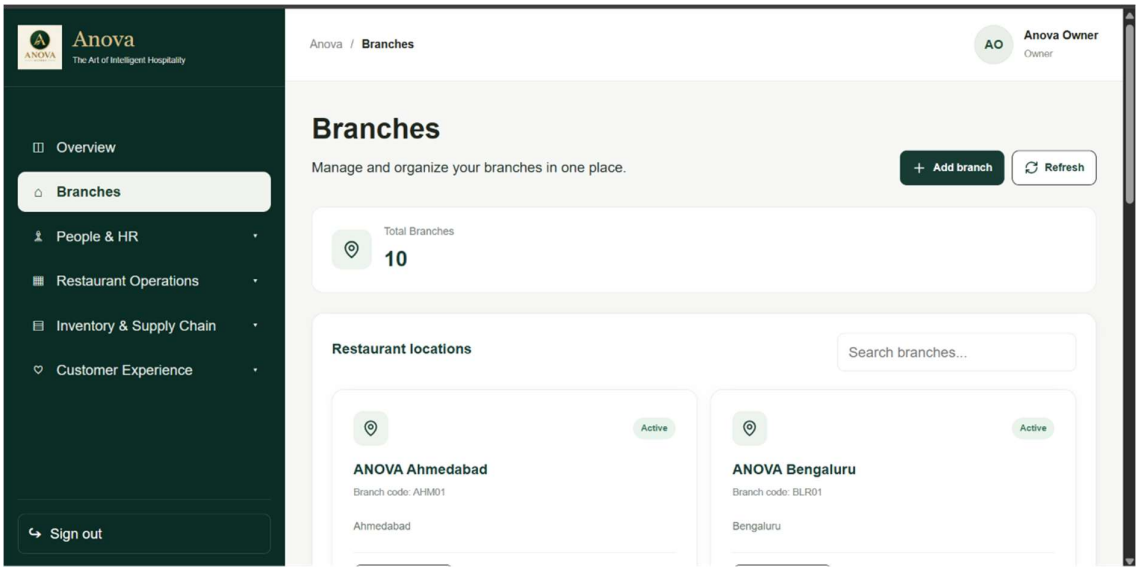



*Figure 10: Branches Page*

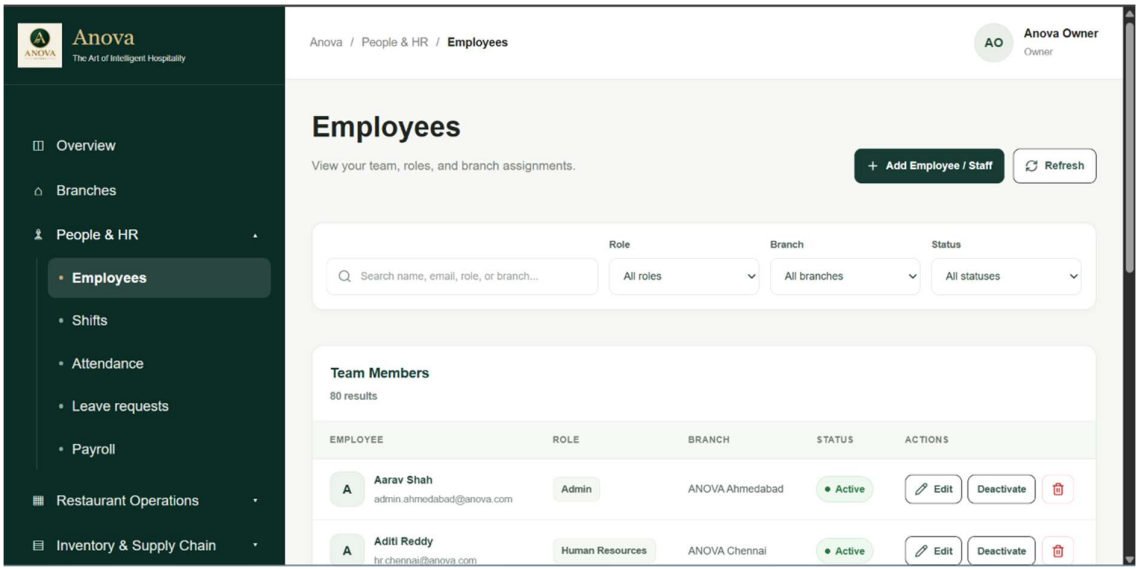



*Figure 11.1: People & HR - Employee Page*

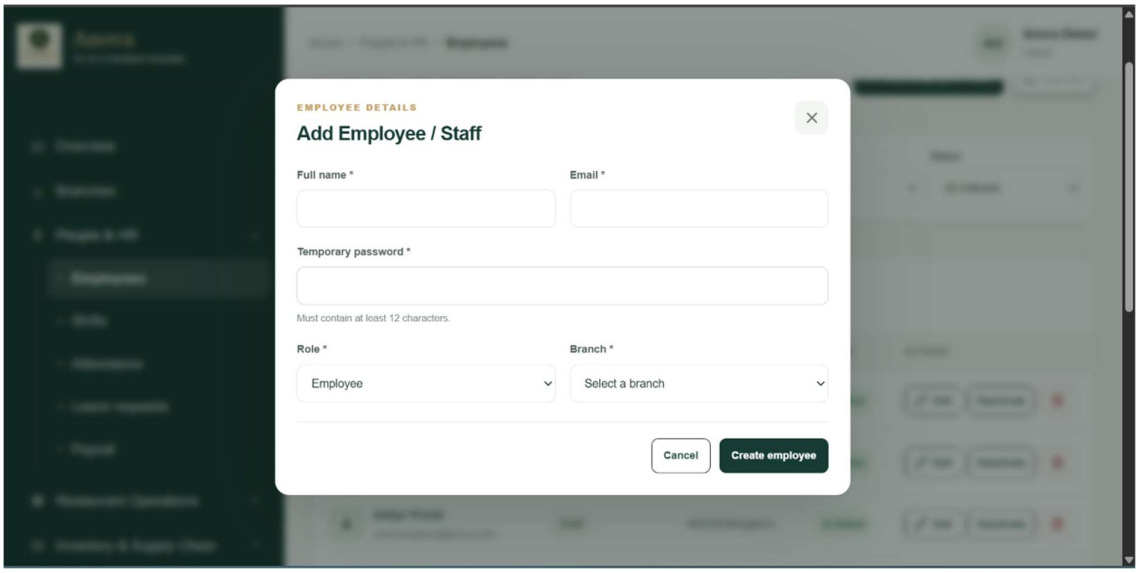



*Figure 11.1.1: Add Employee form*

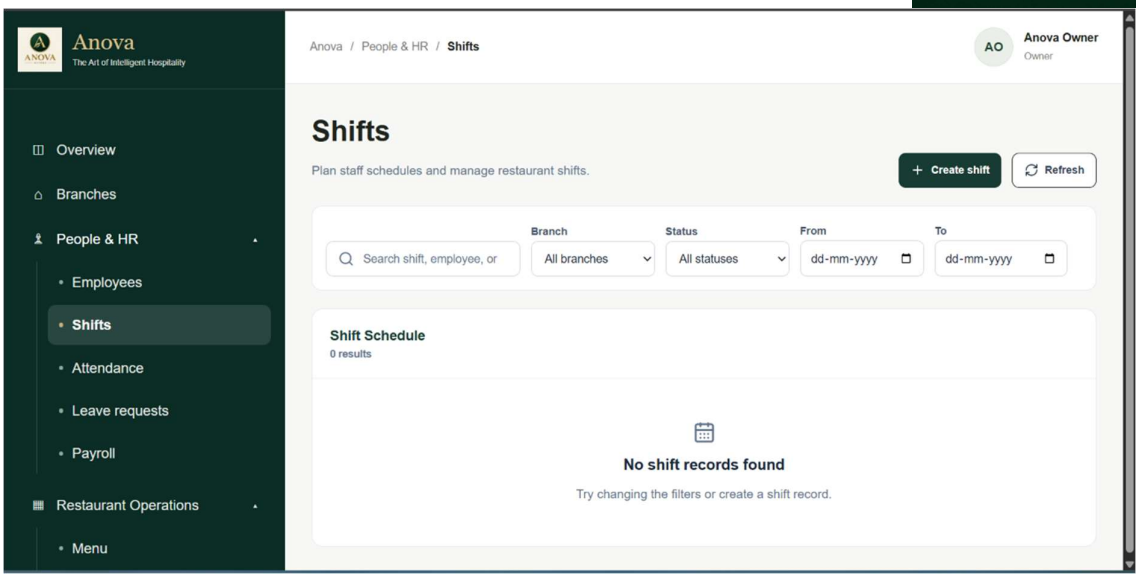



*Figure 11.2: People & HR - Shifts Page*

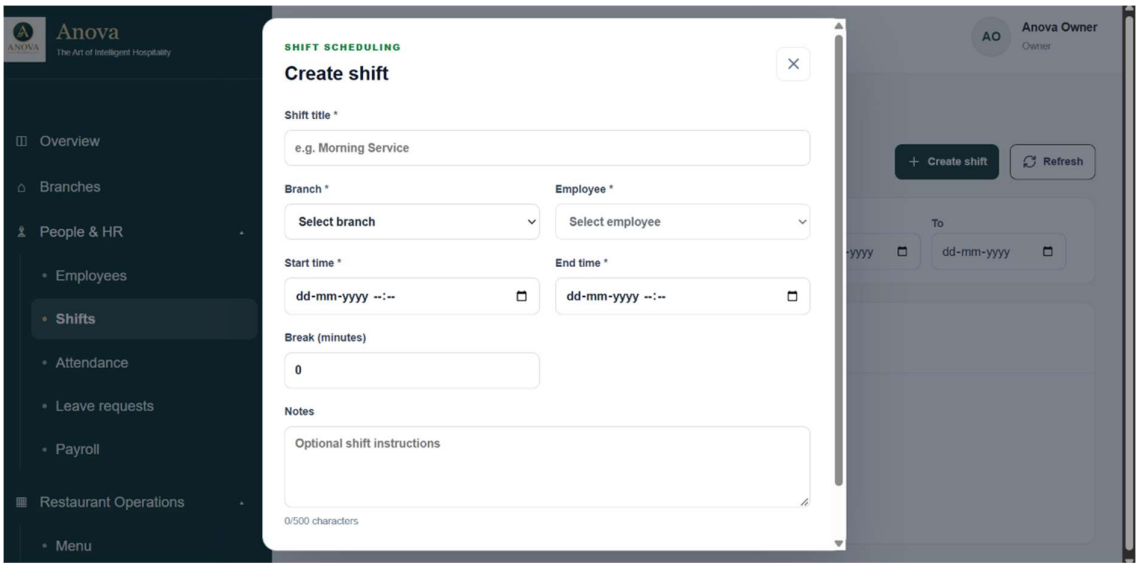



*Figure 11.2.1: Create Shift form*

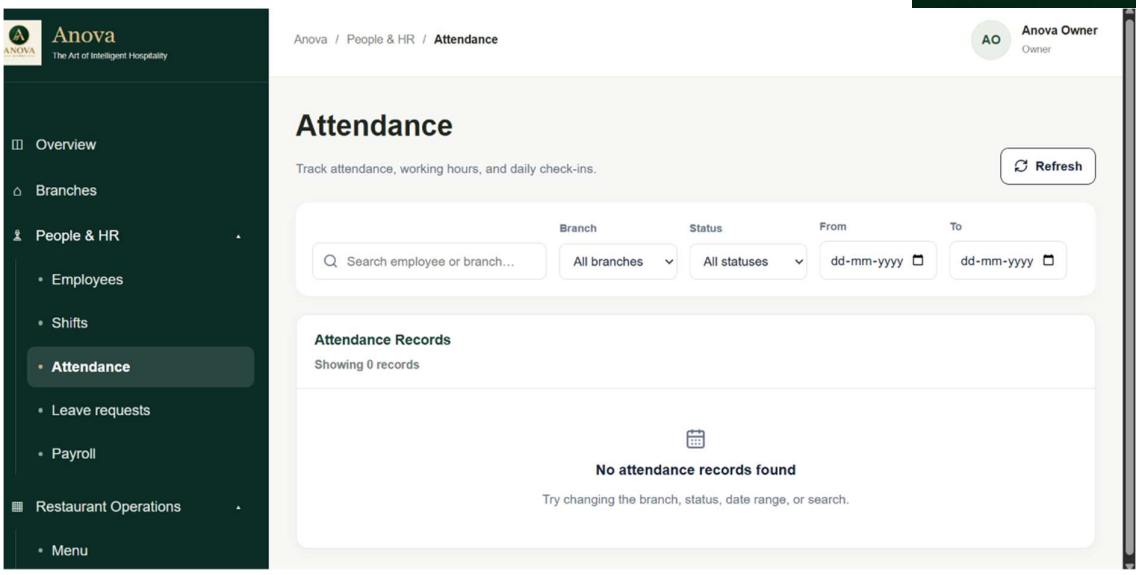



*Figure 11.3: People & HR - Attendance Page*

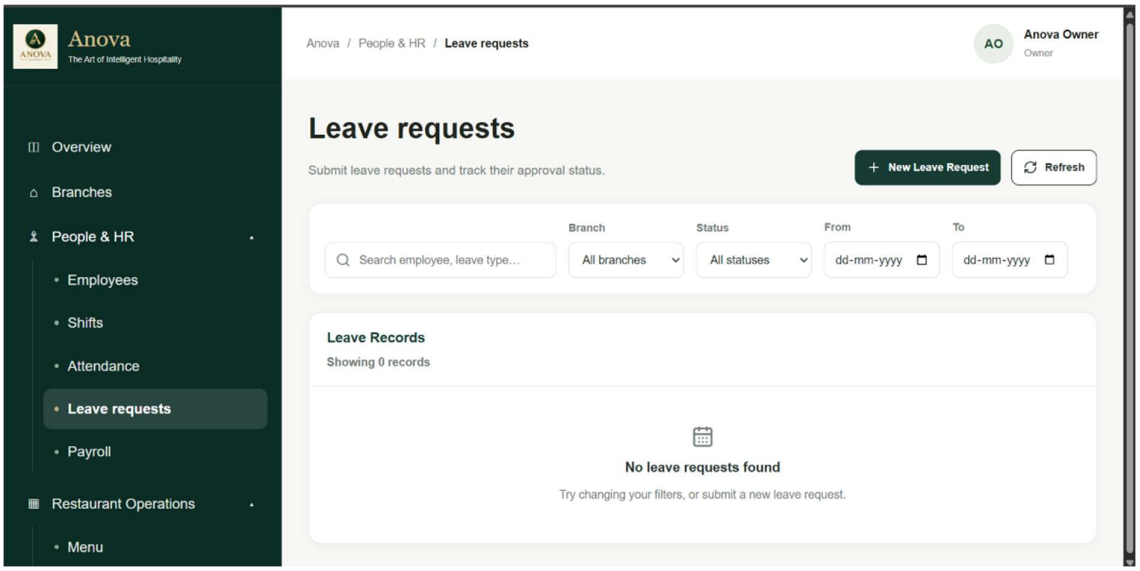



*Figure 11.4: People & HR - Leave Request Page*

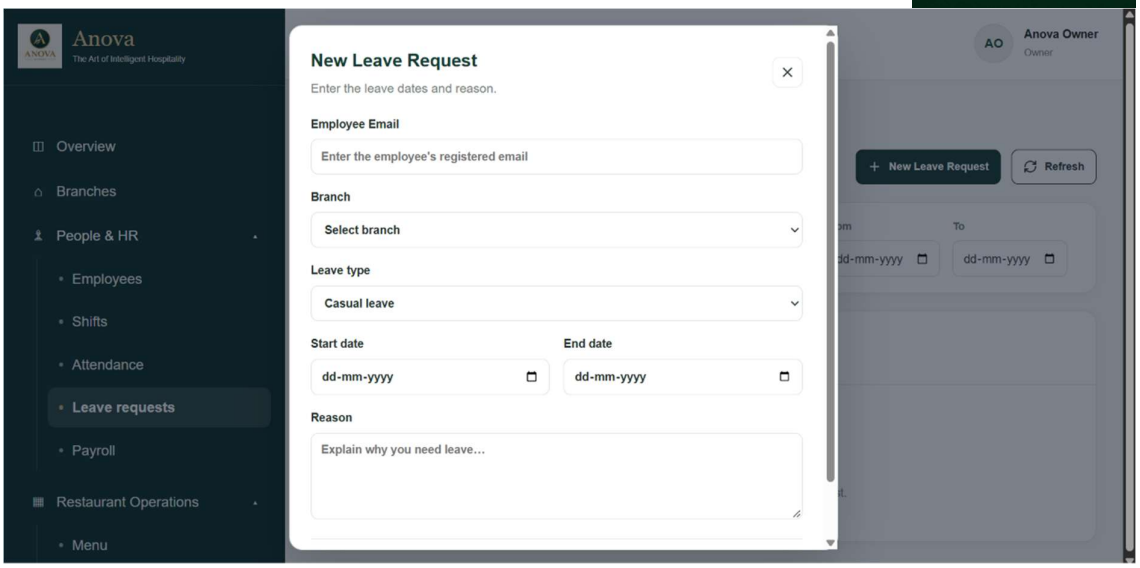



*Figure 11.4.1: Add New Leave Request form*

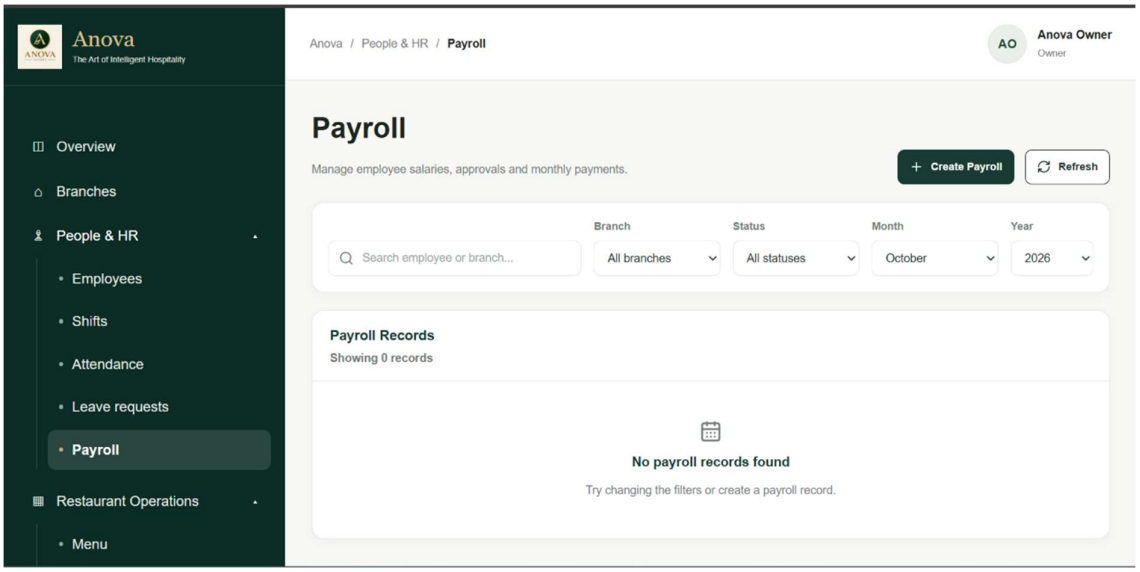



*Figure 11.5: People & HR - Payroll Page*

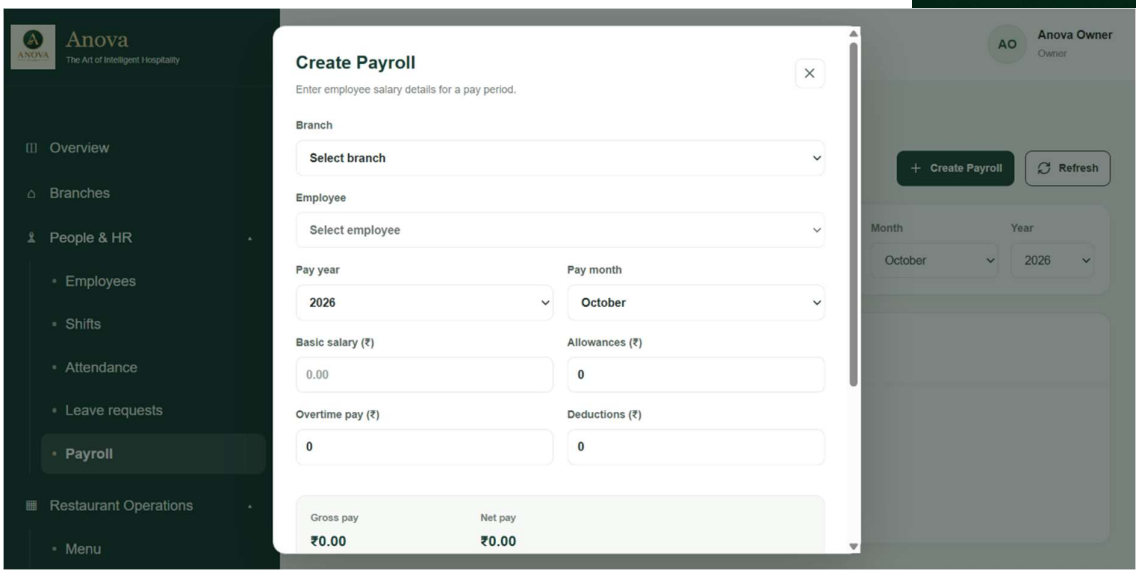



*Figure 11.5.1: Create Payroll form*

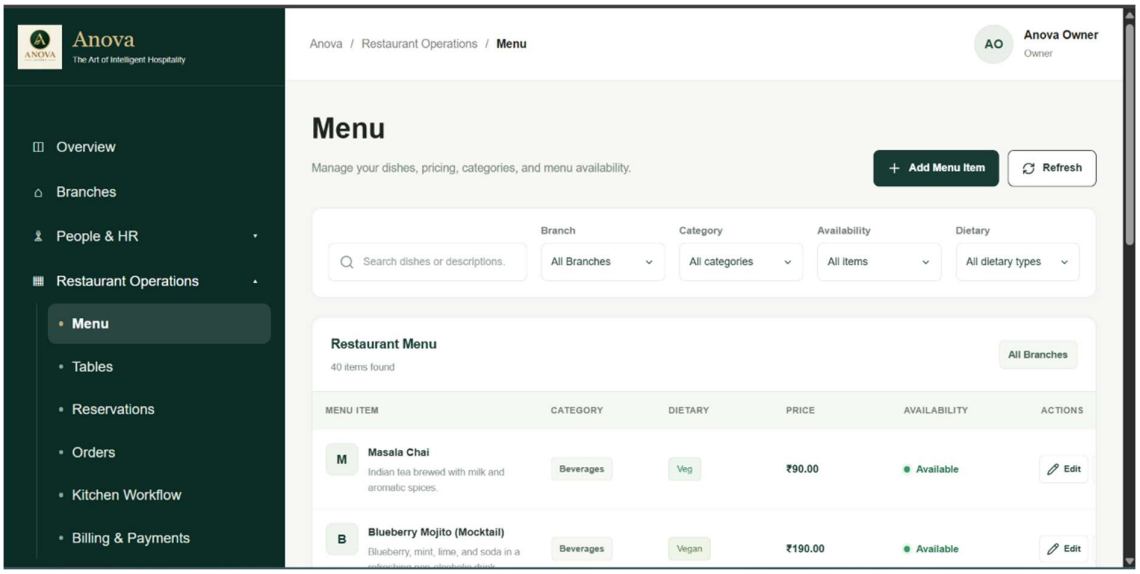



*Figure 12.1: Restaurant Operations - Menu Page*

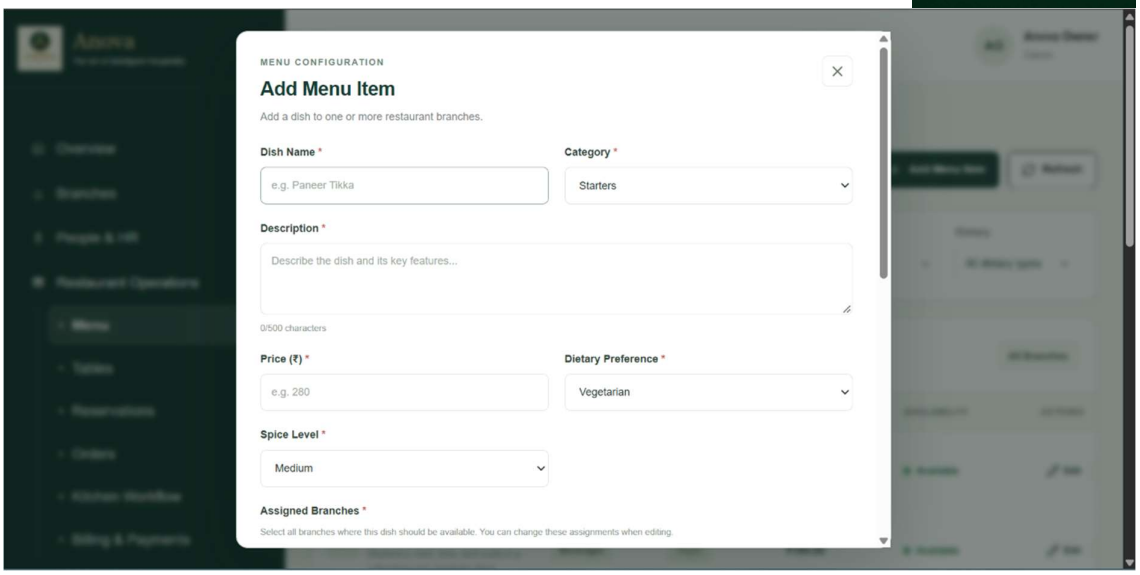



*Figure 12.1.1: Add Menu Item Form*

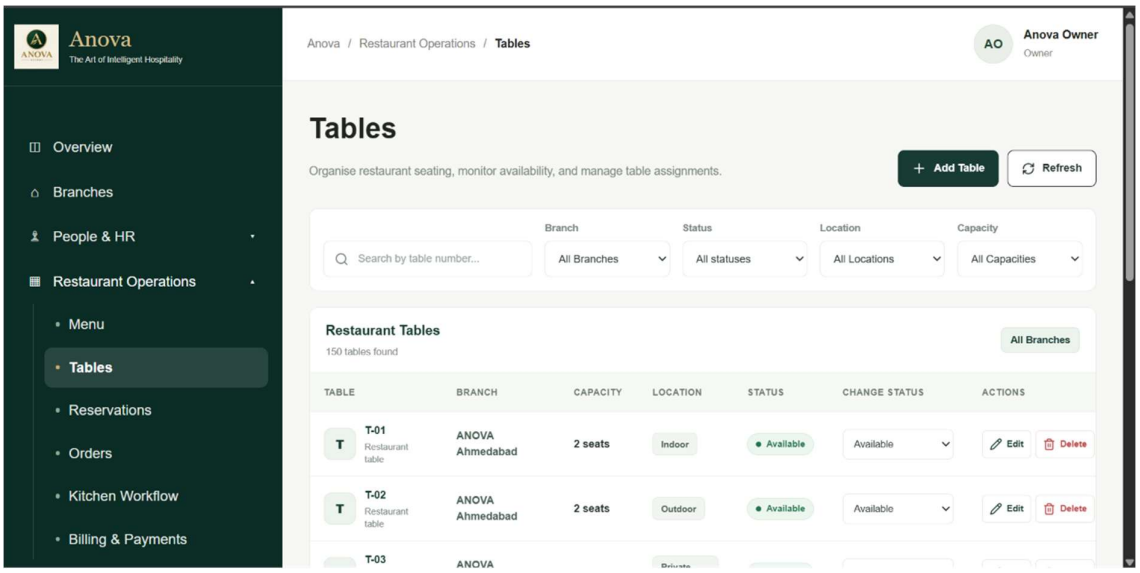



*Figure 12.2: Restaurant Operations - Table Page*

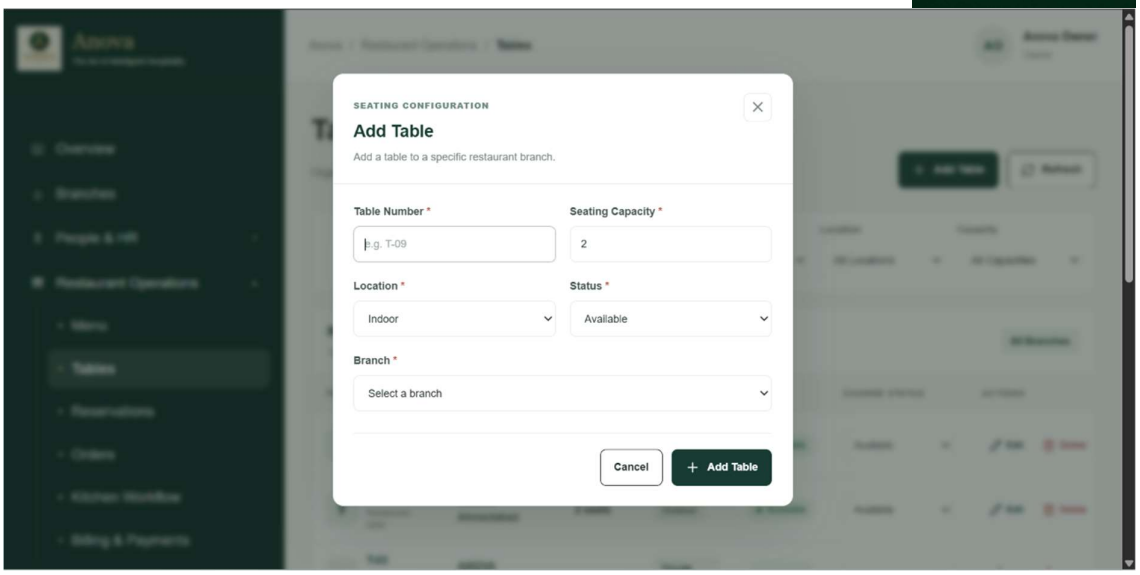



*Figure 12.2.1: Add Table Form*

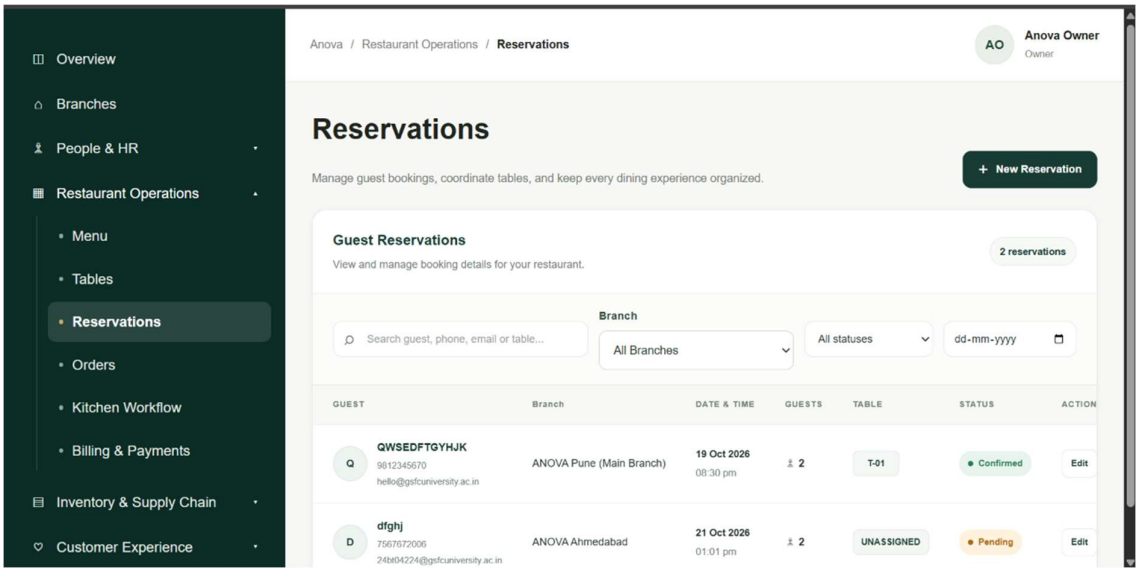



*Figure 12.3: Restaurant Operations - Reservations Page*

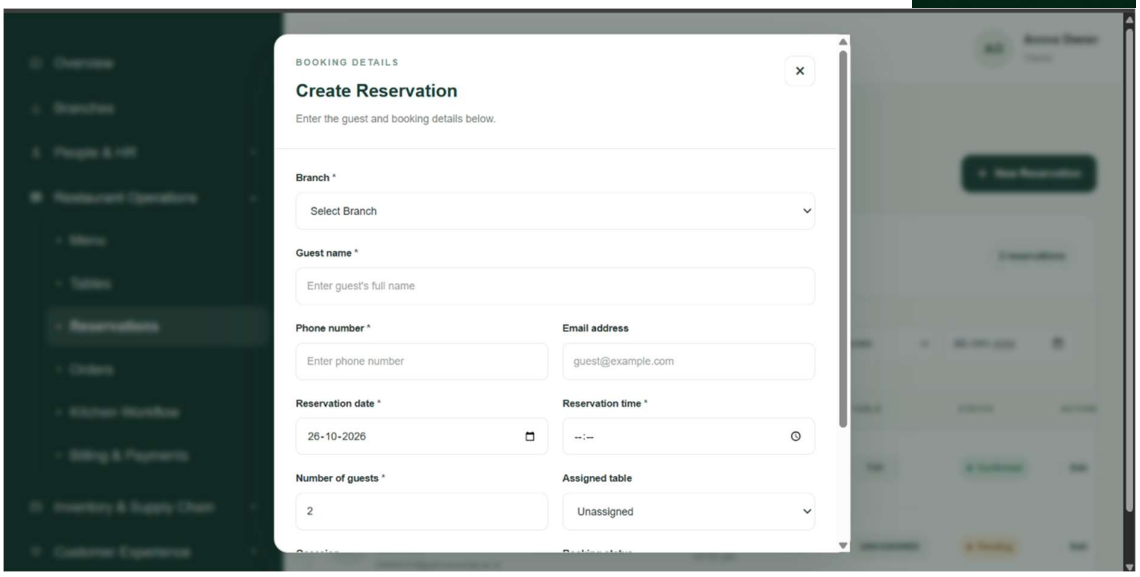



*Figure 12.3.1: Create Reservation Form*

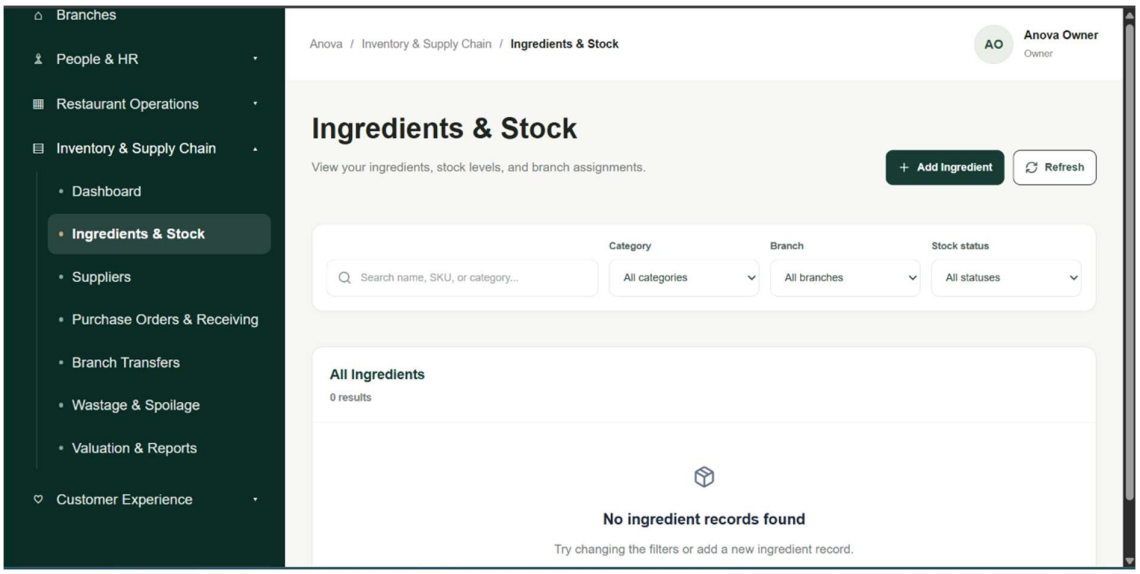



*Figure 13.1: Inventory & Supply Chain - Ingredients & Stock Page*

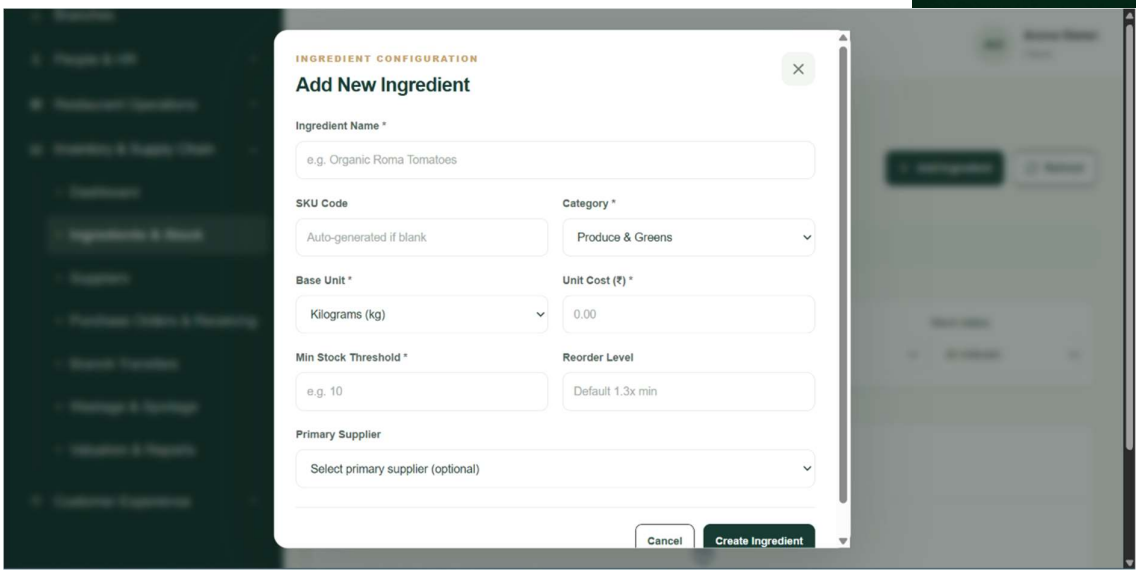



*Figure 13.1.1: Add New Ingredient Form*

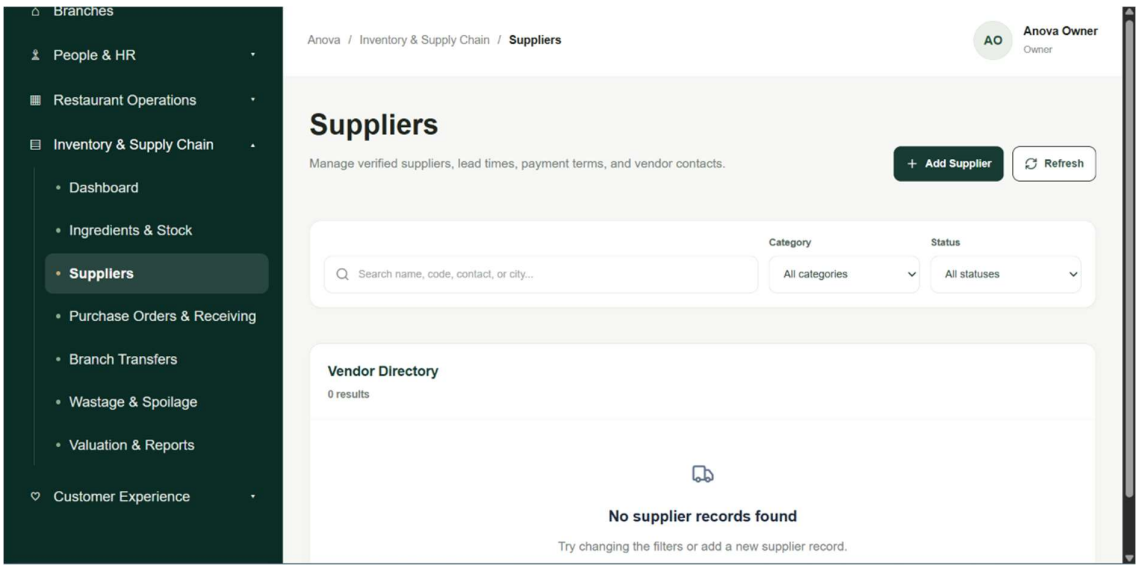



*Figure 13.2: Inventory & Supply Chain - Suppliers Page*

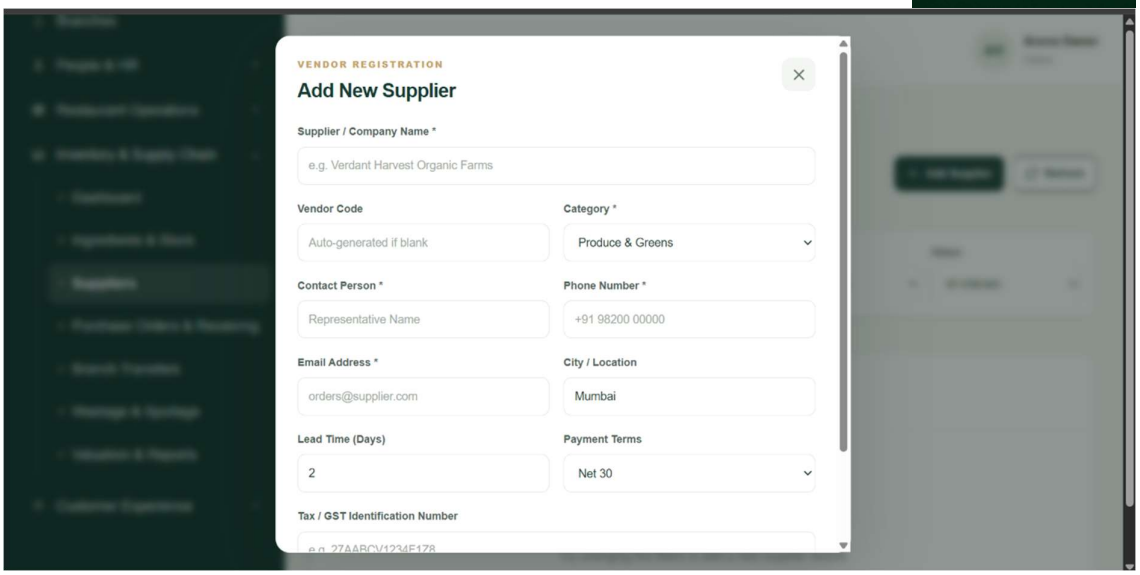



*Figure 13.2.1: Add New Supplier Form*

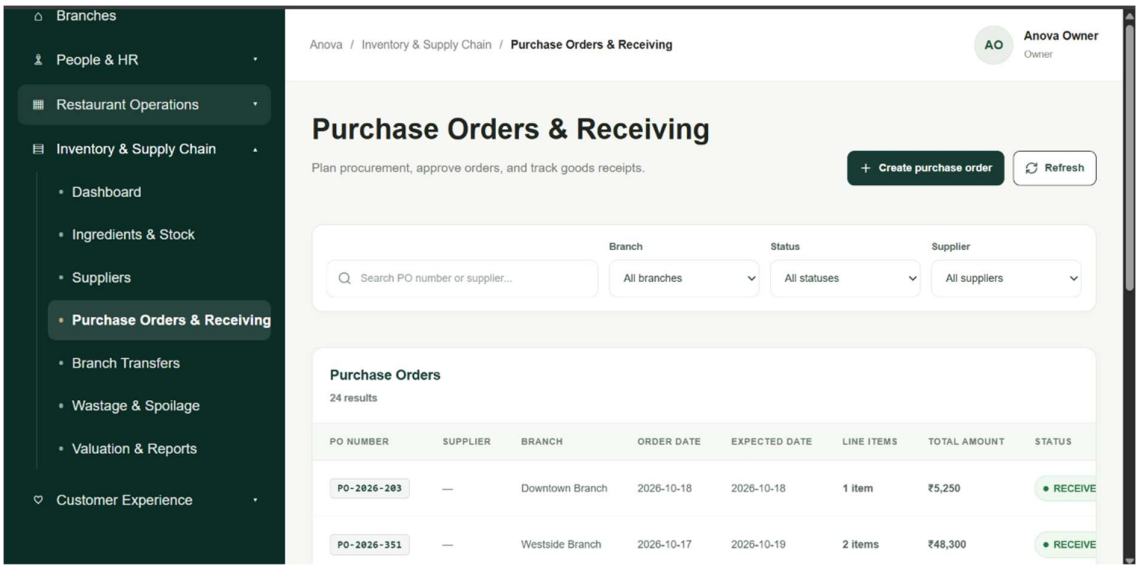



*Figure 13.3: Inventory & Supply Chain - Purchase Orders & Receiving Page*

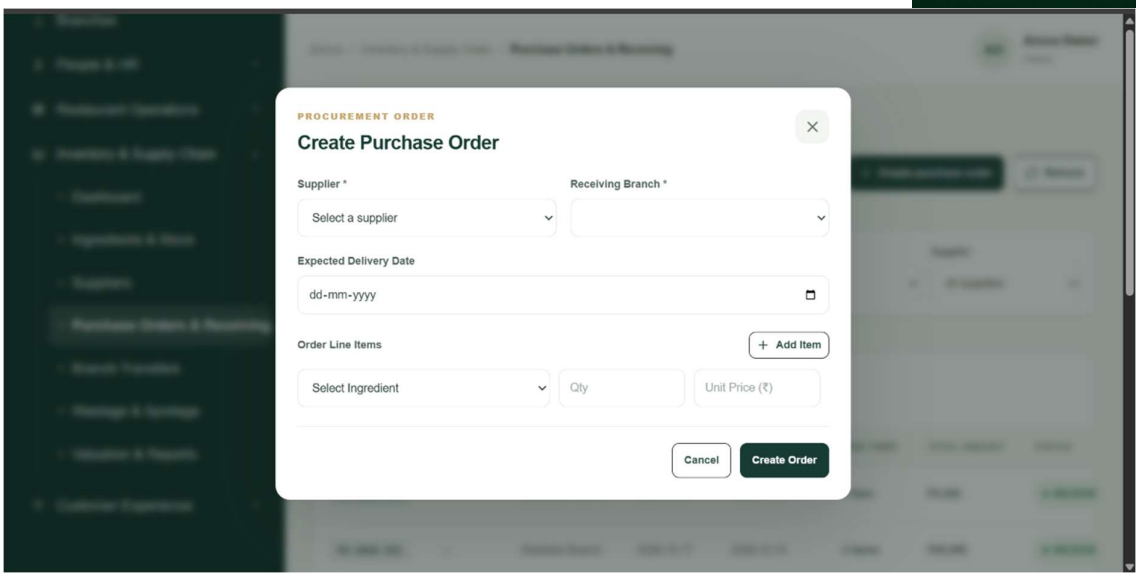



*Figure 13.3.1: Create Purchase Order Form*

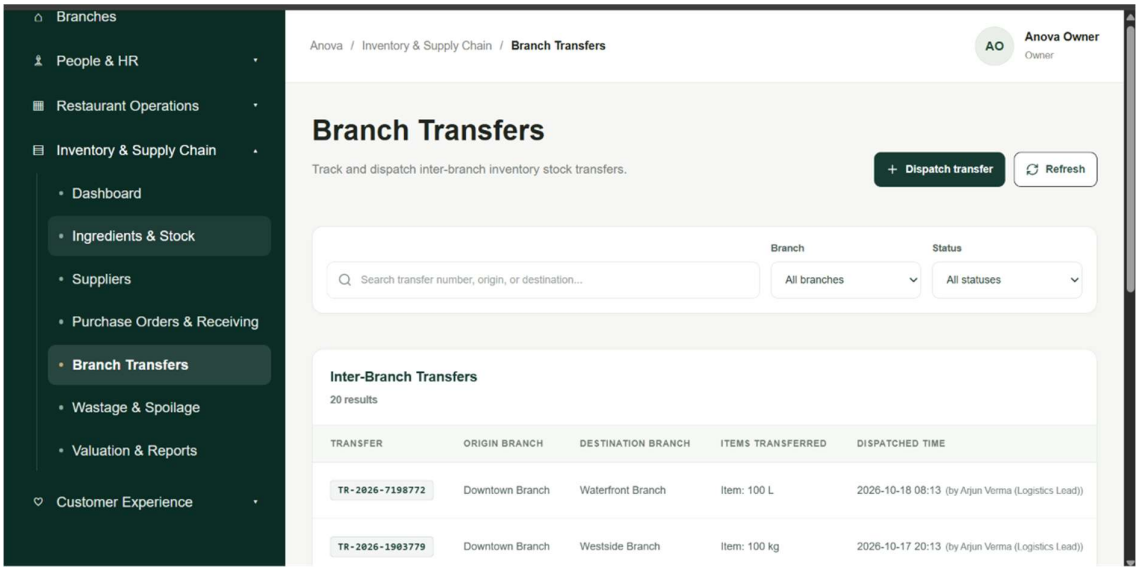



*Figure 13.4: Inventory & Supply Chain - Branch Transfers Page*

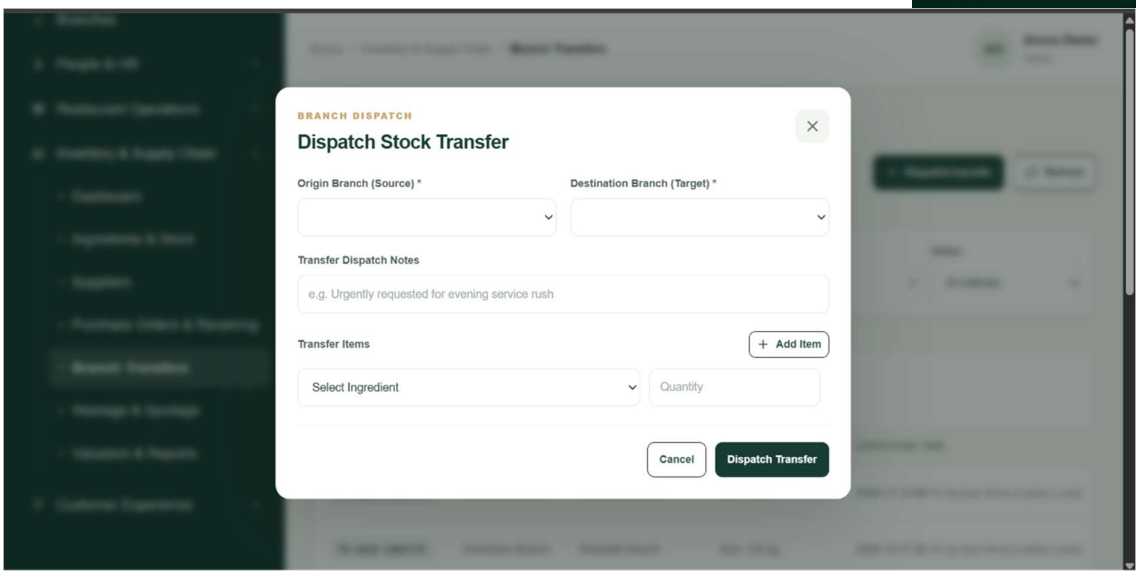



*Figure 13.4.1: Dispatch Stock Transfer Form*

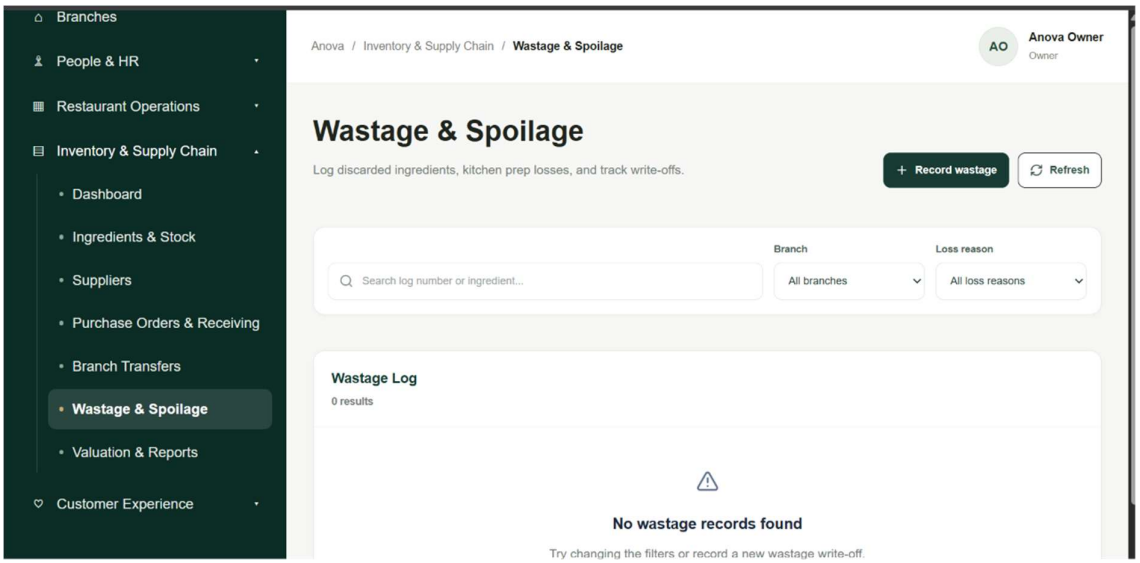



*Figure 13.5: Inventory & Supply Chain - Wastage & Spoilage Page*

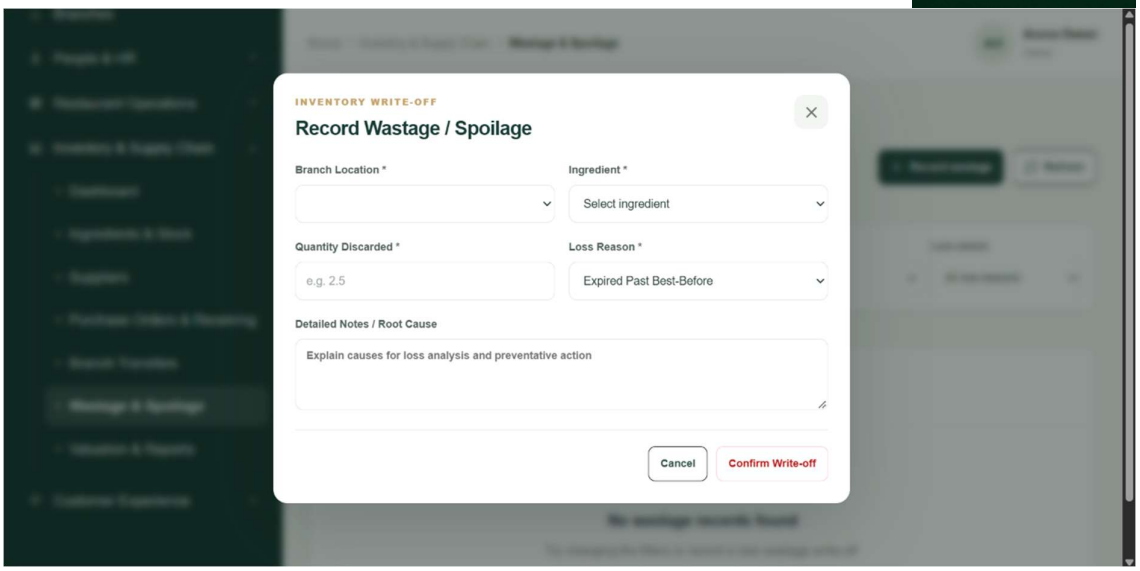



*Figure 13.5.1: Record Wastage/Spoilage Form*

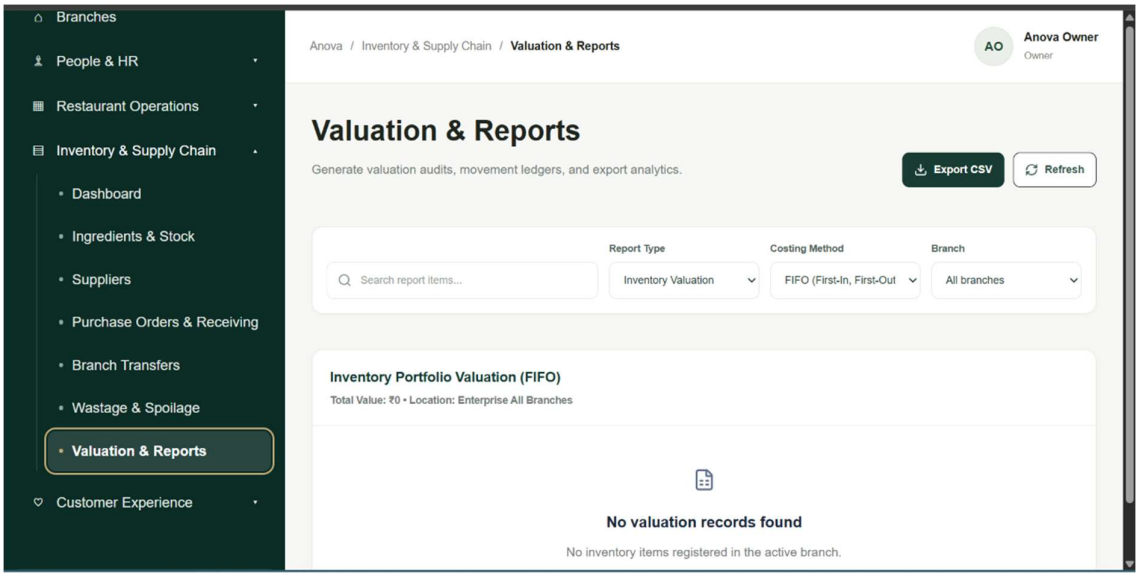



*Figure 13.6:Inventory & Supply Chain - Valuation & Reports Page*

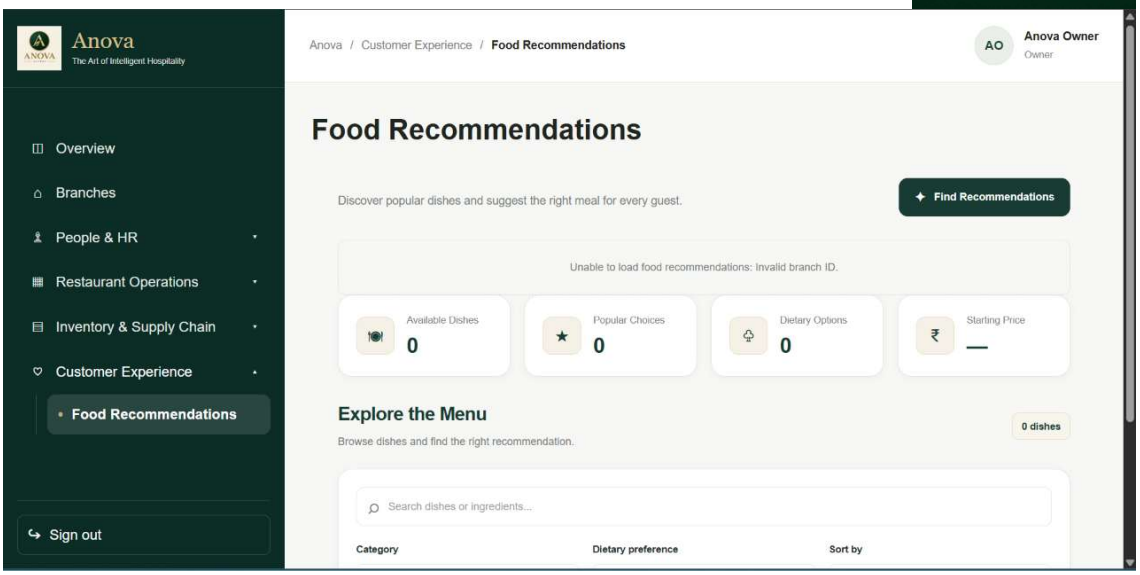



*Figure 14: Customer Experience - Food Recommendations Page*

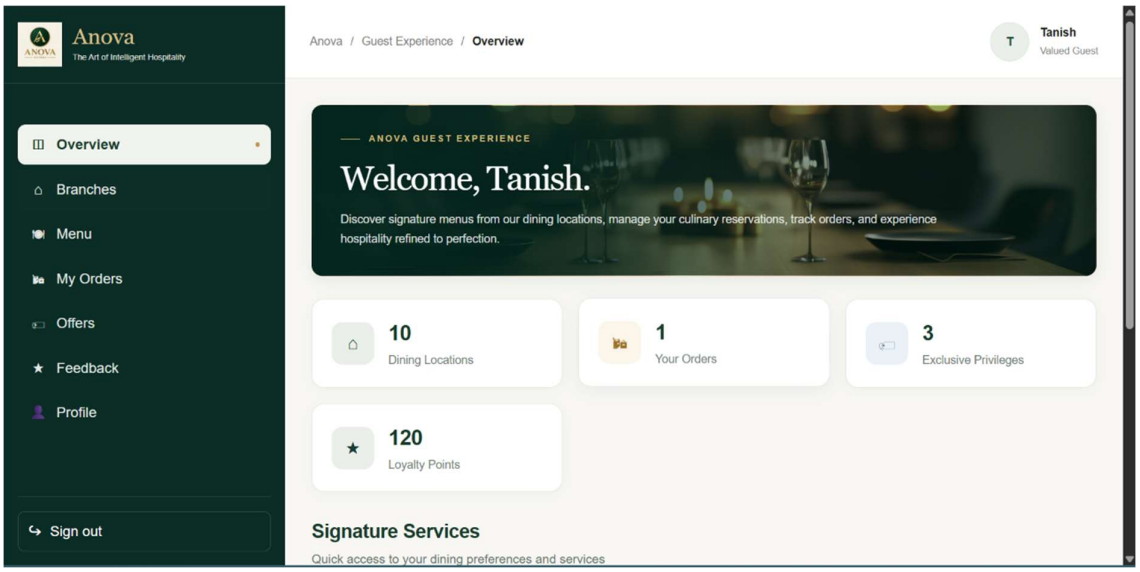



*Figure 15:Customer Dashboard*

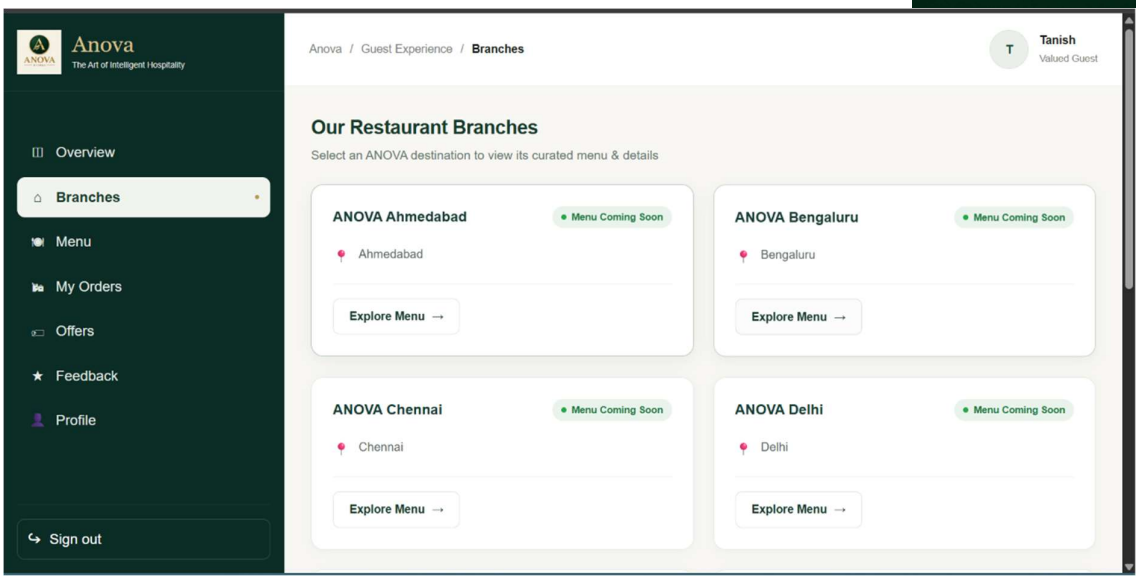



*Figure 16: Branches Page*

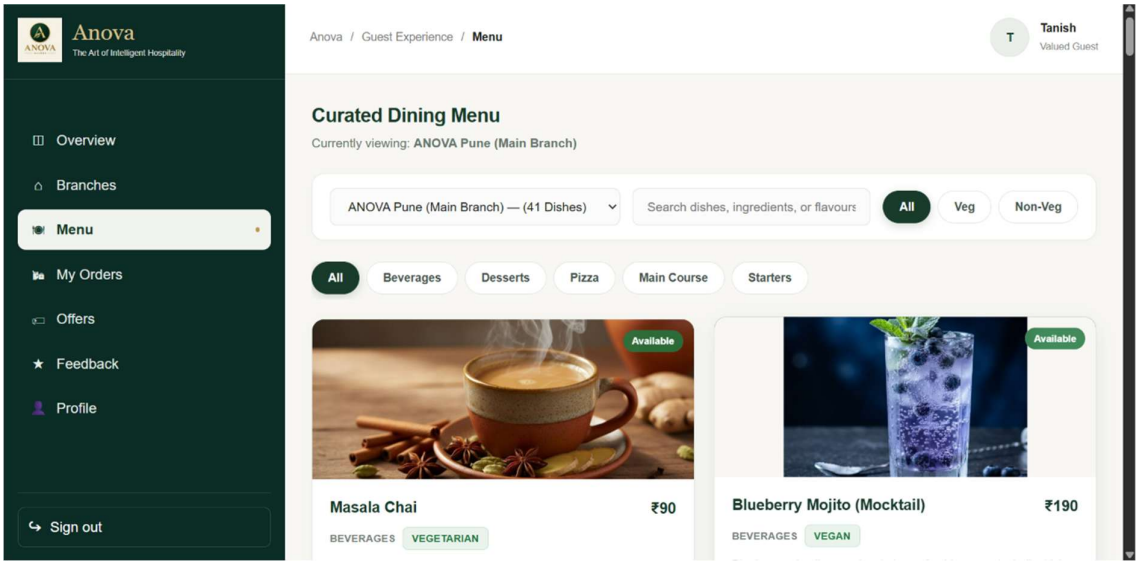



*Figure 17: Menu Page*

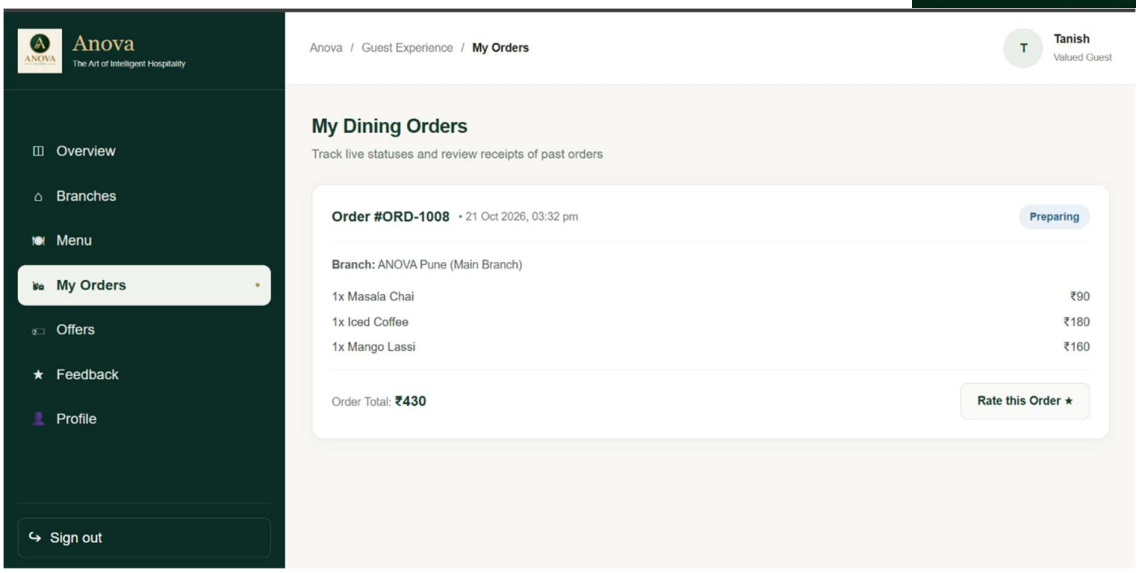



*Figure 18: Orders Page*

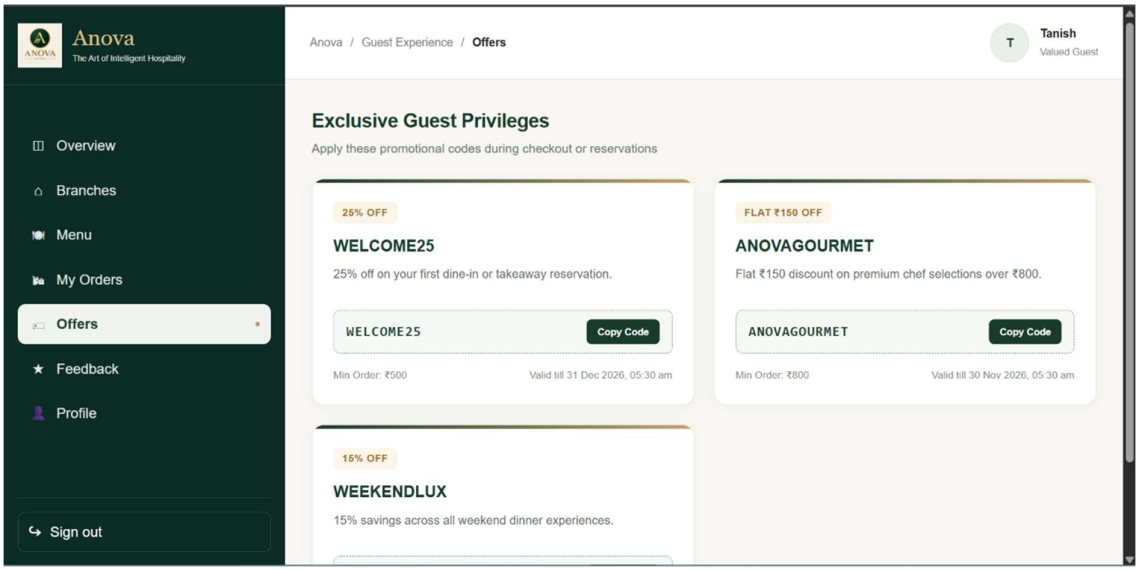



*Figure 19: Offers Page*

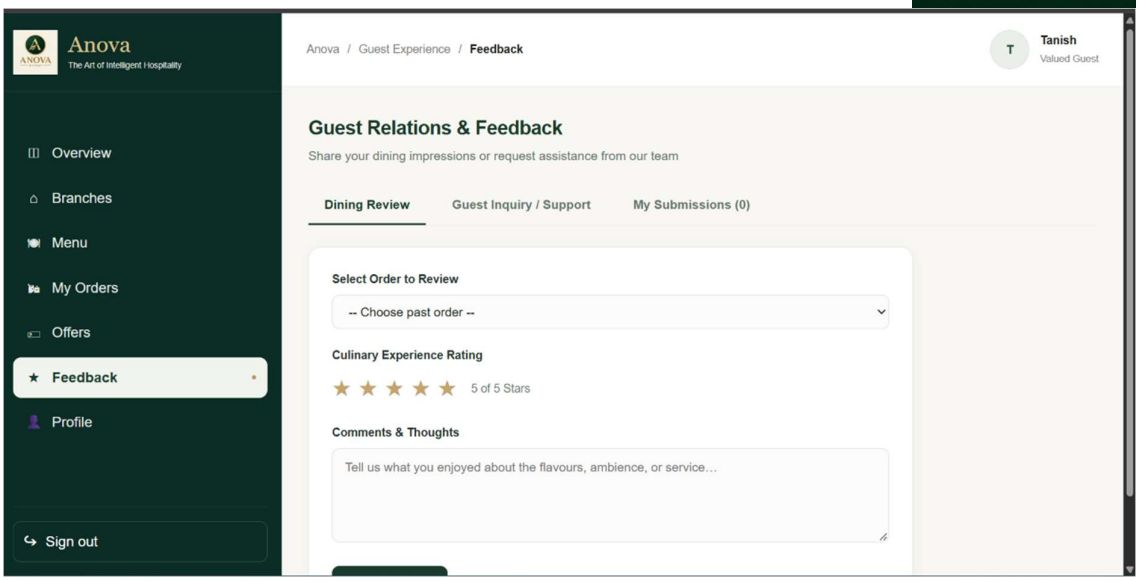



*Figure 20: Guest Relation & Feedback*

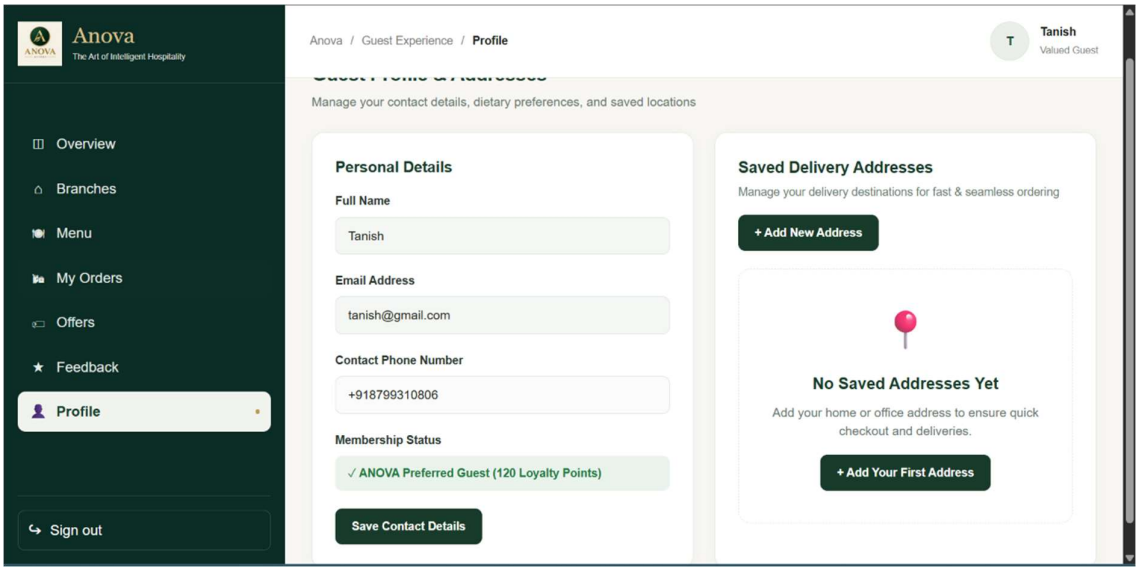



*Figure 21: Customer Profile Page*

# Chapter 4: Observations

## i. Problems Encountered

During the analysis, design, and development of ANOVA – Multi-Branch Restaurant Enterprise Management System, several technical and organizational challenges arise due to the integration of multiple modules into a single web application. Since the system handles different user roles, branch-specific records, and interconnected workflows, careful planning and validation are required to ensure consistency and security.

The major problems encountered during the project are:

- Role-Based Access Control: Managing different user roles and ensuring that each user can access only the features permitted for their role.
- Database Design: Organizing data related to branches, employees, inventory, suppliers, orders, and customers while maintaining data consistency.
- Module Integration: Integrating different modules and ensuring smooth communication between the React frontend, Express.js backend, and MongoDB database.
- Security: Implementing secure authentication, password hashing, and authorization to protect sensitive information.
- Inventory and Order Management: Maintaining accurate stock records and coordinating inventory updates with order processing.
- Time Management: Completing multiple modules within the available time while balancing development, integration, and testing.
- Testing and Debugging: Identifying errors, validating inputs, and ensuring that the system behaves correctly across different modules.

## ii. Lessons Learned

The development of ANOVA provides an opportunity to understand how software engineering concepts are applied to a practical, multi-module web application. It demonstrates the importance of planning, structured implementation, database management, security, integration, and testing throughout the software development lifecycle.

The major lessons learned or expected from the project are:

- Modular Development: Dividing the application into independent modules makes the system easier to develop, test, maintain, and extend.
- MERN Stack Development: Gained a better understanding of how MongoDB, Express.js, React.js, and Node.js work together to build a web application.
- Database Management: Learned the importance of proper database design, data validation, and maintaining consistency across related records.
- Authentication and Security: Understood the importance of JWT-based authentication, password hashing, and role-based access control.
- API Integration: Learned how REST APIs enable communication between the frontend and backend and support smooth data exchange.
- Teamwork and Coordination: Recognized the importance of communication, task distribution, and coordination when developing multiple modules.
- Testing and Debugging: Understood that regular testing helps identify errors early and improves system reliability.
- Time and Project Management: Learned to prioritize essential features, manage development time, and maintain clear project documentation.

# Chapter 5: Conclusion and Future Enhancement

## i. Conclusion

ANOVA – Multi-Branch Restaurant Enterprise Management System is designed to provide a centralized and efficient solution for managing restaurant operations across multiple branches. The system integrates essential functions, including employee management, inventory control, supplier management, menu management, order processing, billing, customer service, and reporting into a unified web-based platform.

Developed using the MERN stack, the project demonstrates the integration of modern web technologies to create a modular, maintainable, and scalable application. Role-based access control helps manage user permissions, while branch-level access supports the organized administration of multiple restaurant locations.

The major outcomes of the project include the consolidation of restaurant operations, improved organization of business data, reduced dependence on manual record-keeping, and better coordination between departments. Inventory and order management features support systematic stock monitoring and order processing, while dashboards and reports provide useful operational insights for decision-making.

The project also demonstrates the practical application of software engineering concepts, including database design, REST API development, authentication, authorization, modular architecture, and system testing. Although features such as live payment processing, GPS-based delivery tracking, and advanced predictive analytics can be added in future versions, the existing system establishes a strong foundation for further enhancement.

In conclusion, ANOVA is more than a restaurant management application; it represents a step toward smarter, more connected, and data-driven restaurant operations. By bringing multiple business functions together within a single platform, the system aims to help restaurant businesses improve efficiency, maintain consistency, and manage growth more effectively. With future enhancements and wider adoption, ANOVA has the potential to evolve into a comprehensive digital management solution that supports the changing needs of modern restaurant enterprises.

## ii. Future Enhancement

The system can be enhanced in the following ways:

- Online Payment Integration: Integrate secure payment gateways to support real-time digital transactions.
- Advanced Analytics: Introduce predictive analytics to forecast sales, customer demand, and inventory requirements.
- Mobile Application: Develop mobile applications for managers, employees, and customers.
- GPS-Based Delivery Tracking: Integrate live GPS tracking to monitor delivery personnel and provide accurate delivery updates.
- Automated Notifications: Add email, SMS, or push notifications for orders, reservations, stock alerts, and employee updates.
- Accounting Integration: Connect the system with accounting software to automate financial records and reporting.
- AI-Based Recommendations: Improve food recommendations using customer preferences, order history, and purchasing patterns.
- Cloud Scalability: Enhance cloud infrastructure to support more branches, users, and transactions efficiently.